The Broken Model Reads Fairest: Head Collapse Conceals Demographic Disparity in Continually Fine-Tuned Encoders

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Abstract

The impact of continual fine-tuning is well understood to be far more pronounced on a model's classification head than on its encoder, and refitting a linear head on the frozen encoder recovers most of the lost accuracy. However, comparatively little attention has focused on how this head collapse affects fairness evaluation. This paper demonstrates that head collapse from recency bias can conceal group disparities by shrinking the set of professions the head predicts, which drives the averaged equalized-odds gap toward zero. On Bias in Bios, a 25-profession class-incremental sequence learned by a BERT-base encoder with an MLP head, the four of nine methods that collapse onto the most recent professions appear to be the fairest (equalized-odds gap ≤ 0.026). A standard fairness table therefore ranks broken models first. Refitting the head on each method's retained replay memory brings the best continual learner to within 0.016 balanced accuracy of joint training. Refitting on a held-out labelled sample of the same size grows the collapsed methods' equalized-odds gaps by a factor of 5.3–8.0, and all nine methods, joint training included, then fall between 0.149 and 0.169. We conclude that the disparity was present all along, in the task and the representation, and that head collapse only hid it. We therefore recommend reporting a model's prediction-level fairness metrics only if it reaches a stated accuracy floor, alongside a readout the trained head cannot corrupt. Finally, we propose three criteria for determining when a reduction in disparity from fairness-aware methods genuinely reflects underlying fairness improvement.

1 Introduction

Pretrained models are rarely trained only once. Rather, they are often continually fine-tuned as new data, classes, and domains arrive [Shi et al., 2024]. Typically, updates are judged mainly by whether they hurt accuracy [Wang et al., 2024]. When the label space grows, this process has a well-documented failure mode: the classifier becomes biased toward classes it trained on most recently, a phenomenon known as task-recency bias [Masana et al., 2023]. Notably, the bias largely affects the classifier head, while the encoder beneath it retains most of what it learned [Rebuffi et al., 2017, Lanzillotta et al., 2026].

The fairness of such models is usually assessed with prediction-level criteria such as equalized odds [Hardt et al., 2016], which compare error rates across demographic groups. At a minimum, one would expect a model with a smaller gap to treat groups more equally. We show that under task-recency collapse this expectation fails.

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On Bias in Bios [De-Arteaga et al., 2019], a 25-profession class-incremental sequence, four of the nine methods we compare collapse onto the five most recent professions. These methods report equalized-odds gaps of at most 0.026, whereas every method above a basic accuracy floor reports between 0.146 and 0.157. A standard fairness table therefore ranks the least accurate models as the fairest.

The small gaps reflect the prediction distribution, not equal treatment. A profession the head never predicts has true- and false-positive rates of zero in both groups, so it contributes nothing to a gap averaged over all 25 professions. Refitting a linear head on frozen features makes the hidden disparity visible. Refitting on each method's retained replay memory brings the best continual learner to within 0.016 balanced accuracy of joint training. With a held-out labelled sample of the same size, the collapsed methods' gaps grow 5.3–$8.0\times$. All nine methods then fall between 0.149 and 0.169, including joint training (0.157), which has no sequence to forget. The disparity is a property of the task and the representation, and the collapsed head only concealed it.

This creates a specific hazard for deployed systems. Accuracy monitoring flags the collapsed checkpoint while fairness monitoring passes it, and the standard repair for the first failure exposes the second. To an operator, fixing an accuracy regression appears to introduce a fairness regression (Sec. 4.2).

Contributions.

1. We show that task-recency collapse drives prediction-level fairness metrics toward zero and inverts the fairness ranking of nine methods, and we remove the inversion with a utility floor below which deployed-head fairness is not reported (Sec. 4.1).
2. We sweep the labelled repair budget from zero to replay size and show the concealed disparity returning with accuracy, with every method converging on the same gap (Sec. 4.2).
3. We apply the same checks to fairness-aware methods, including two of our own that constrain each update to shrink the gap between the worst- and best-off groups' losses. On Bias in Bios the checks remove their apparent advantage, and the loss gap they optimize does not track equalized odds. On a synthetic sequence built to pass the checks, our constraint reduces the DER++ accuracy-parity gap by a factor of 4.3 (Sec. 4.3).

Scope. We study a BERT-base encoder on supervised classification sequences and do not evaluate parameter-efficient adaptation or agentic and embodied settings. We expect the mechanism to transfer but not the specific numbers (Sec. 5). Collapse also does not require encoder drift: with BERT frozen, FAIR-GEM's head still collapses and the refit still recovers it (Appendix G.24). Code and results are available at `https://github.com/MlsBli/broken-model-reads-fairest`.

2 Related Work

Encoders versus classifier heads in class-incremental learning. Previous work in class-incremental learning establishes that sequential training degrades the classifier head far more than the encoder beneath it. The head overpredicts newer classes because of the imbalanced new:old data ratio. Methods that target this bias can restore old-class accuracy by normalizing or rescaling the final layer [Wu et al., 2019, Hou et al., 2019, Zhao et al., 2020], while linear probes (that fit a fresh linear classifier on the frozen encoder) demonstrate that the encoder's features still separate old classes [Davari et al., 2022, Lanzillotta et al., 2026]. Accordingly, replacing the final layer, by refitting it or by classifying with class means, restores overall accuracy [Rebuffi et al., 2017, Lanzillotta et al., 2026] and, under spurious correlations, worst-group accuracy [Kirichenko et al., 2023]. While prior work measures accuracy alone, we investigate the impact of this head–encoder asymmetry on group-fairness evaluation. To reveal disparities hidden by collapsed heads, we retrain the last layer.

Parity by degradation. The fairness literature notes that group parity can be achieved by degrading a model rather than by treating groups more equitably. Criteria that compare a rate across groups, such as equalized odds [Hardt et al., 2016], are satisfied trivially by a constant predictor (always predicting "yes" gives every group the same true- and false-positive rates). In practice, fairness methods often close gaps by lowering performance for the best-off group rather than raising it for the worst-off [Zietlow et al., 2022, Mittelstadt et al., 2024], and Mittelstadt et al. [2024] propose minimum performance thresholds in place of strict parity. In our setting, the degradation comes from

ordinary class-incremental training rather than a fairness objective. Our utility floor applies the same principle to evaluation rather than training, so a model cannot be counted as fair unless it reaches a minimum accuracy.

Distinctions between representation and prediction bias. NLP fairness research has illustrated that bias in a model's internal representations and bias in its predictions are distinct and can diverge. Intrinsic metrics computed on embeddings do not predict downstream bias [Goldfarb-Tarrant et al., 2021], and debiasing a model can leave the protected attribute recoverable from its encoder [Elazar and Goldberg, 2018, Orgad et al., 2022]. Here, head collapse creates the same divergence without any debiasing. Although the collapsed heads report almost no gap, a gender probe recovers gender well above chance on every encoder we train (Appendix G.27). We therefore report each deployed-head gap alongside the gap measured on a refitted head.

3 Experimental Setup

Datasets. Bias in Bios [De-Arteaga et al., 2019] is a 25-profession class-incremental dataset over biography texts. It contains five professions per task, with binary gender as the protected attribute. Spurious Shift is our novel six-task domain-incremental synthetic dataset, built following Park et al. [2026]. We use it only to demonstrate the utility of our newly introduced fairness methods. Both are described in Appendix A.

Model. Every method fine-tunes a BERT-base encoder with an MLP classification head. Logits of classes not yet introduced are masked. Full training details are in Appendix B.1.

Methods compared. We compare nine methods: joint retraining across all data (an oracle), fine-tuning, A-GEM [Chaudhry et al., 2019], DER++ [Buzzega et al., 2020], GroupDRO [Sagawa et al., 2020], a FaIRL-inspired adversarial variant [Basu Roy Chowdhury and Chaturvedi, 2023], FSW [Park et al., 2026], and our two novel methods, FAIR-GEM and Fair-DER++ (Appendix B.2). Ours, marked $\star$, add one constraint to a backbone. Each takes the gradient of the gap between the worst and best group's loss on the current batch (Eq. 2) and requires the update to decrease the gap by a dynamic margin (Eq. 3). FAIR-GEM alternates this projection with A-GEM's memory projection, and Fair-DER++ applies it once to the DER++ update (Eq. 4). Replay methods carry an episodic memory of 800 samples per past task, stratified by group and equal across methods. Methods with additional parameters were tuned on a shared validation objective that rewards balanced accuracy and penalises the accuracy-parity gap (Eq. 1).

Evaluation. We report balanced accuracy, the accuracy-parity gap, and the equalized-odds gap. Equalized odds is traditionally defined for binary labels, so on Bias in Bios (profession labeling) we compute it for each profession against the rest and combine the results across professions (Appendix B.3). Each encoder is read three ways: through its deployed head, through a linear head refitted on frozen features, and through training-free nearest-class-mean and nearest-neighbour rules. Results are mean $\pm$ sd over ten seeds.

4 Results

Measured on the deployed head, prediction-level fairness metrics reward continual learners for collapsing. As a result, the least accurate models are rated the fairest (Sec. 4.1) and refitting the head illuminates the cause. As accuracy improves, the fairness disparity also increases, and all methods converge to the joint retrain disparity (Sec. 4.2). This is innate task-related disparity rather than learner-related disparity. Therefore, fairness-aware methods do not demonstrate benefits over standard replay methods (although fairness methods do cut disparity on a sequence built to add it, Sec. 4.3). Each result suggests an inexpensive check to evaluate continual learners: gate fairness on accuracy, refit the head before trusting gaps, and compare against joint retraining before applying a fairness intervention.

4.1 Collapsed Models Report the Smallest Gaps

Misleadingly, models with collapsed accuracy report the smallest fairness gaps (Figure 1, Table 1). Fine-tune, A-GEM, GroupDRO and FAIR-GEM predict only the five professions of the final task, which limits their balanced accuracy to about 0.20, yet their equalized-odds gaps are at most 0.026. The smallest gap among functioning models is 0.146, at 0.740 balanced accuracy. A fairness ranking of this comparison therefore places the four least accurate models first.

Table 1: **On the deployed head four methods collapse, and a refit head recovers all of them.** Head: deployed head. Mem: head refit on retained memory. $\Delta_{\mathrm{EO,head}}$ is shown only above 0.5 balanced accuracy. †: joint training. ⋆: ours. Bold: best continual learner. Mean ± SD over ten seeds.

Method	$\mathrm{BalAcc}_{\mathrm{mem}}$ ↑	$\mathrm{BalAcc}_{\mathrm{head}}$ ↑	$\mathrm{Worst\text{-}grp}_{\mathrm{mem}}$ ↑	$\Delta_{\mathrm{EO,head}}$ ↓
Joint retrain †	.748±.003	.740±.001	.695±.003	.146±.003
Fine-tune	.704±.004	.193±.014	.650±.003	—
A-GEM	.721±.004	.205±.025	.672±.007	—
DER++	.700±.005	.651±.023	.650±.007	**.146**±.009
GroupDRO	.703±.004	.191±.017	.650±.004	—
FaIRL-insp.	.602±.007	.381±.030	.558±.007	—
FSW	.679±.004	.650±.013	.629±.007	.149±.007
FAIR-GEM ⋆	**.732**±.003	.207±.025	**.683**±.005	—
Fair-DER++ ⋆	.719±.004	**.659**±.009	.666±.006	.157±.008

In practice, the small fairness gaps are a result of the collapsed heads only predicting recent professions rather than true parity. The equalized-odds gap measures the difference between true- and false-positive rates across groups. When a model never predicts a profession, the rates are zero and the gap is zero, contributing to lowering the average equalized-odds gap.

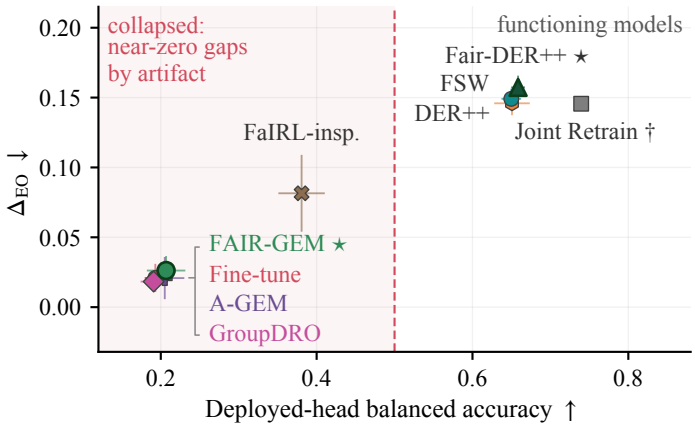


Figure 1: **Collapsed models report the smallest fairness gaps.** Deployed-head equalized-odds gap against balanced accuracy on Bias in Bios. Below the dashed utility floor, the collapsed models score as fairest. Error bars: ±1 SD over ten seeds.

The inversion is not specific to equalized odds. Accuracy parity, a max–min over per-group macro-recall rather than an RMS over one-vs-rest rate gaps, inverts further, with the collapsed methods at 0.006–0.009 against 0.063–0.067 for the functioning ones. Any criterion that compares rates across groups inherits it.

Because the metrics themselves cannot reveal the problem, we gate them on accuracy. Deployed-head fairness is reported only for models above a balanced-accuracy floor of 0.5, and excluded entries are marked rather than printed. Every floor in $[0.39, 0.64]$ admits the same 3 methods, so the rule does not depend on where the floor is set. A deployed-head fairness gap is interpretable only when the head still works, and one accuracy comparison is enough to check.

4.2 Repairing the Head Brings the Gap Back

A small gap on a collapsed head has two potential explanations: either the model treats the groups more equally, or the collapse makes the disparity unmeasurable. To investigate which is occurring, we refit a linear head on the frozen encoder, which repairs the head and leaves the representation unchanged. We sweep the labelled budget B from zero to the size of the replay buffer each method

already stores, to trace how accuracy and disparity recover together (limiting B at the replay buffer so that even the largest budget costs no more labelled data than replay already uses). To standardize, the budget comes from a held-out split shared across methods.

The evidence supports the second explanation (Figure 2). As the refitted head recovers accuracy, the disparity reappears. With a budget the size of the replay buffer, A-GEM's balanced accuracy rises from 0.205 to 0.707, and its equalized-odds gap rises from 0.025 to 0.169. Across the four collapsed methods, the gap grows 5.3–8.0-fold.

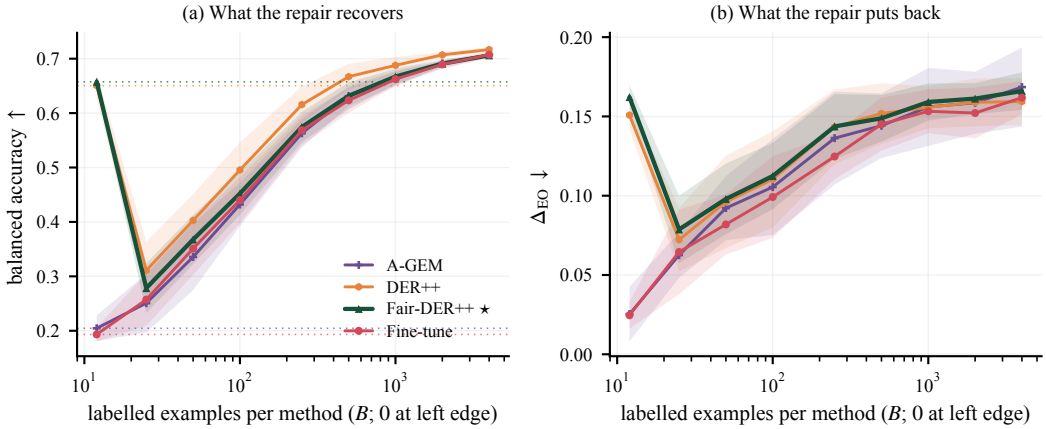


Figure 2: **Repairing the head restores accuracy and the gap collapse was hiding.** A linear head refit on B labelled examples, from the deployed head ($B = 0$) to replay-buffer size. (a) Balanced accuracy, with dotted lines at each deployed head. (b) Equalized-odds gap. Bands: ± 1 SD over ten seeds.

At replay size every method converges on the same gap, between 0.149 and 0.169. Joint retraining, which sees all five tasks at once and has nothing to forget, lies inside this range at 0.157. The disparity is therefore inherited from the task and the representation, not introduced by continual learning.

The collapse is a failure of the head alone. With BERT frozen, FAIR-GEM still collapses (0.184 balanced accuracy) and the refitted head still recovers it (0.603, Appendix G.24), so refitting the head, not retraining the model, is enough to restore it.

This creates a trap for deployed systems that monitor accuracy and fairness separately. A-GEM is a concrete case. After the final task, its accuracy monitor correctly flags a broken model, while its fairness monitor reports one of the smallest gaps in the comparison. Refitting the head, the standard fix, restores its accuracy and raises its gap to the level of every other method, so the repair appears to have introduced a fairness regression. The disparity was in fact present all along, hidden by the collapsed head. Refitting the head therefore serves as a fairness diagnostic as well as a repair.

4.3 The Same Checks Decide Whether a Fairness Method Works

Fairness-aware continual learners are judged with the same prediction-level metrics, so they inherit the same failure. On the deployed head, two of them rank among the fairest models in Table 1: GroupDRO reports an equalized-odds gap of 0.018 and FAIR-GEM 0.026, against 0.146 for DER++. Both have collapsed, so the table credits them with the artifact of Sec. 4.1. A smaller gap is evidence for a fairness method only if three checks pass. The head must still work, the readout must be one the head cannot corrupt, and the quantity the method optimizes must track the metric being reported.

On Bias in Bios, the first two checks remove the apparent gain. Above the utility floor, no fairness-aware method improves on DER++ (Table 1). With a refit head, all nine methods fall within 0.020 of one another (Sec. 4.2): FAIR-GEM reaches 0.169 and GroupDRO 0.160, against 0.159 for DER++ and 0.157 for joint retraining. Accuracy parity on a shared refit pool shows the same tie (Figure 3a).

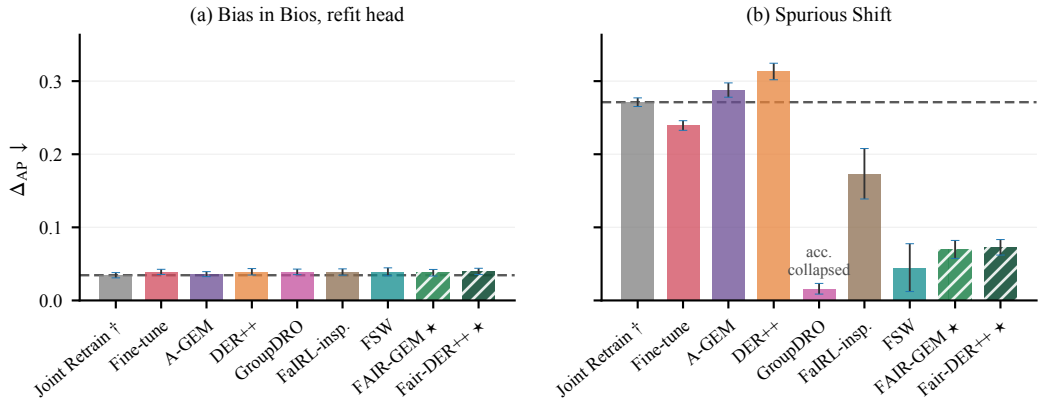


Figure 3: **The constraint lowers the gap only where the metric can register it.** Accuracy-parity gap per method, with joint training dashed. (a) Bias in Bios, measured with a head refit on a shared pool. (b) Spurious Shift. Hatched: ours. Error bars: ± 1 SD over ten seeds.

The third check suggests why the constraint has no effect. Both of our methods shrink the gap between the worst and best group's mean loss, pooled over all professions (Eq. 2). On Bias in Bios this quantity does not track equalized odds. Its rank correlation with the reported gap is 0.002 across all methods, seeds and tasks. Excluding collapsed models raises it to 0.476, and conditioning the loss gap on the label raises it to 0.695 (Appendix D.2). Collapse therefore decouples the training signal from the metric, as it decouples the metric from the model, and the pooled loss gap differs from a label-conditional criterion even without collapse.

Spurious Shift is built to pass all three checks. Its tasks share one binary label space, so no head can collapse onto recent classes, and labels are balanced within each group, so the pooled loss gap tracks equalized odds closely ($\rho = 0.938$). A shortcut feature that grows more predictive for the majority group makes disparity rise over the sequence, and DER++ ends at $\Delta_{\mathrm{AP}} = 0.313$, above joint retraining at 0.271 (Figure 3b; Appendix A). Here the constraint works. Fair-DER++ cuts the DER++ gap by a factor of 4.3, to 0.073 ($p = 0.002$, 10 paired seeds), with the best accuracy retention of any method (Acc-BWT $= -0.011$, Appendix C).

The first check still applies without collapse. GroupDRO reports the smallest gap on Spurious Shift ($\Delta_{\mathrm{AP}} = 0.016$) at 0.532 balanced accuracy, barely above chance on a binary task. This is the same parity by degradation, and a fixed floor of 0.5 would admit it. The floor has to be set relative to chance, which is 0.04 on Bias in Bios and 0.5 here.

The two benchmarks therefore do not disagree about the constraint. They differ in whether the reported metric can register its effect, and the three checks identify which case a benchmark is in before any method is credited.

5 Discussion and Conclusion

Task-recency collapse makes a continually trained classifier look fair. A collapsed head predicts few labels, so prediction-level fairness metrics shrink toward zero, and the least accurate models in our comparison report the smallest gaps. Refitting the head restores accuracy and disparity together, and every method, joint retraining included, converges on the same gap. Nothing here implies that fairness metrics are uninformative in continual learning. The claim is narrower: a fairness number read from a deployed head is evidence about the model only when that head still works.

Recommendations. First, report deployed-head fairness only above an accuracy floor set relative to chance. On Bias in Bios a floor of 0.5 suffices, but on a binary task the same floor admits a near-chance model (Sec. 4.3). Second, report at least one readout the trained head cannot corrupt, such as a refitted linear head, and more than one when the result is a ranking, since refit and nearest-class-mean readouts order the encoders differently (Appendix D.1). Third, before crediting a fairness-aware

method, check that the quantity it optimizes tracks the metric being reported, and compare its disparity with that of joint retraining to see how much the sequence added.

Beyond this setting. The mechanism depends only on how many labels the head predicts, not on the encoder or its scale, and the floor is a threshold on an accuracy that every deployment already computes. How often continually updated foundation-model classifiers collapse is a separate empirical question, and we do not answer it. The conditions for concealment are common, however. Pretrained models are routinely adapted in place [Shi et al., 2024], the diagnostics that gate those updates are almost entirely accuracy-based [Wang et al., 2024, Luo et al., 2023], and nothing in such a pipeline would surface the concealment. Qi et al. [2024] report a similar pattern for safety alignment, which erodes under fine-tuning that leaves task performance intact. In both cases a headline metric says little about a property it does not measure.

Limitations. Our evidence comes from one real benchmark, a BERT-base encoder and a binary protected attribute, and we do not test larger backbones, parameter-efficient adaptation or agentic settings. The explanation for the null result on Bias in Bios rests on correlations between the optimized loss gap and the reported metric rather than on an intervention. Remaining limitations are in Appendix G.2.

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A Benchmarks

Spurious Shift is a six-task synthetic sequence built to separate overall task accuracy from demographic fairness. A rotating feature block forces accuracy forgetting, while an independent "shortcut" feature grows in predictiveness for the majority group so a model can retain accuracy while its minority-group performance decays. Each task draws 4,000 training, 2,000 validation and 10,000 test examples with a binary protected attribute at 20% minority prevalence. The construction follows Park et al. [2026], and the full generative process is in Appendix G.5.

Bias in Bios [De-Arteaga et al., 2019] is a 25-profession class-incremental sequence over biography text, five professions per task, with binary gender as the protected attribute. It serves a different role: its expanding output space entangles representation retention, classifier recency, demographic drift, and surrogate alignment, so we treat it as a multiclass stress test rather than a clean identification benchmark. Appendix G.27 separates these effects and Appendix G.10 gives preprocessing details.

Both provide (x, y, g) per example, with g available to fairness-aware methods during training and to every method at evaluation.

B Methods

B.1 Setup

Each task t provides training data $\{(\mathbf{x}_t^{(i)}, y_t^{(i)}, g_t^{(i)})\}_{i=1}^{n_t}$ with sensitive group g. An episodic memory $\mathcal{M}$ has a capacity of 800 samples per past task, stratified by group (Appendix G.9). The realized allocation is asserted equal across all replay methods at runtime. All Spurious Shift results are mean $\pm$ sd over ten seeds and all Bias in Bios results over ten seeds. Methods with benchmark-specific free parameters are selected with Optuna. After the full task sequence, each trial is scored by the uniform task average

$$Q_{\text{val}} = \frac{1}{T}\sum_{t=1}^{T}\left(\text{BalAcc}_{\text{val},t} - 0.75\,\Delta_{\text{AP},\text{val},t}\right), \tag{1}$$

which is maximized without a secondary metric or Pareto-stage tie-break. The 0.75 weight is fixed and identical for every method, so it cannot favour one over another. The test set is never used for selection. Bias in Bios uses 20 trials per method and a 10k-sample-per-task cap; Spurious Shift uses 15 trials and all available training data. Methods without free parameters share the fixed benchmark training configuration. Both frozen manifests include FSW.

Spurious Shift. All methods share a 3-layer MLP (256 hidden, BatchNorm, dropout 0.1) trained with AdamW (lr 3×10^{-4}, weight decay 10^{-4}), 3-epoch linear warmup with cosine annealing, gradient clipping at norm 3.0, batch size 512, and early stopping (patience 5, δ=10^{-3}).

BIOS. Each method uses a BertMLP: a BERT-base encoder ($\sim$110M params) with a 3-layer MLP classification head (Appendix G.11). A two-group AdamW optimizer applies $\sim2\times10^{-5}$ to BERT and 3×10^{-4} to the head, with the per-method BERT learning rate tuned independently. Training runs 2 epochs per task with step-level warmup (6% of total steps). Gradient projection operates on the full $\sim$110M-parameter space. Unseen-class logits are masked to $-\infty$ before the softmax, so classes do not enter the denominator before they are introduced. Once introduced, new and old classes compete normally.

B.2 Baselines

We compare seven methods under a matched memory budget and the tuning protocol of Section B.1: **joint retraining** on all data seen so far, an oracle upper bound marked †; **fine-tuning** with no mitigation; the continual learners **A-GEM** [Chaudhry et al., 2019] and **DER++** [Buzzega et al., 2020]; the static fairness methods **GroupDRO** [Sagawa et al., 2020] and a **FaIRL-inspired** adversarial variant [Basu Roy Chowdhury and Chaturvedi, 2023]; and **FSW** [Park et al., 2026], the closest prior fairness-aware continual learner, which reweights current-task samples by a per-epoch linear program. Full descriptions, including the domain-incremental adaptation FSW required, are in Appendix G.31.

B.3 Metrics

We build a lower-triangular result matrix, so after each task t all methods are evaluated on tasks $0, \ldots, t$. We report:

- **Balanced accuracy (BalAcc) ↑ :** macro-average recall across the classes present in an evaluation task.
- **Accuracy-parity gap (AP) ↓ :** $\Delta_{\mathrm{AP}} = \max_g \mathrm{BalAcc}_g - \min_g \mathrm{BalAcc}_g$.
- **Equalized-odds gap (EO) ↓ :** on binary Spurious Shift, the maximum of the between-group TPR and FPR gaps. On multiclass Bias in Bios, the maximum of their one-vs-rest RMS gaps across professions.
- **Backward transfer (BWT) ↑ :** $\frac{1}{T-1}\sum_{i=0}^{T-2}(R_{T-1,i} - R_{i,i})$, where $R_{j,i}$ is BalAcc on task i after training through task j. Negative values indicate forgetting.
- **Fairness backward transfer (F-BWT) ↑ :** $\frac{1}{T-1}\sum_{i=0}^{T-2}(F_{i,i} - F_{T-1,i})$, where F is the AP-gap matrix. Negative F-BWT means fairness on past tasks worsened.
- **Demographic-parity gap (DP) ↓ :** the between-group positive-prediction-rate gap on binary Spurious Shift. On Bias in Bios, the sum of the absolute per-profession prediction-rate differences.

(Arrows indicate the direction of improvement.) For AP, DP, and worst-group accuracy, groups with fewer than 100 test samples (20 on Bias in Bios) are excluded. EO and TPR additionally require each eligible class–group cell to meet that threshold. Model selection uses validation data only, and the test set is held out until final evaluation.

C Spurious Shift: Full Results

On a synthetic sequence built to isolate demographic drift, adding a fairness constraint does cut the accuracy-parity gap by a factor of 4.3 ($p = 0.002$, 10 paired seeds), but it does not transfer in practice. This section gives the full results that Sec. 4.3 summarises.

Accuracy forgetting. Fine-tune exhibits severe forgetting under the feature rotation (Acc-BWT $= -0.416 \pm 0.006$), and GroupDRO behaves similarly because it has no retention mechanism. Fair-DER++ has the best observed Acc-BWT (-0.011 ± 0.002), compared with -0.035 for DER++ and -0.016 for the joint-training reference. Projection without loss-level replay is less effective here, as A-GEM reaches only -0.342. FSW combines sample reweighting with τ-weighted replay and reaches Acc-BWT $= -0.058 \pm 0.014$, although it does so at a lower final accuracy (Figure 7a; Appendix G.13).

Fairness forgetting. Accuracy retention does not imply fairness retention. DER++, which is the strongest baseline on final accuracy, has the largest AP gap ($\Delta_{\mathrm{AP}} = 0.313 \pm 0.011$), and this is consistent with replay preserving demographic shortcuts alongside predictive features. Fine-tune and A-GEM also accumulate large gaps (0.239–0.288). GroupDRO reaches a small gap (0.016) only with severe accuracy collapse, and a loss-level penalty on the same disparity signal reaches a comparable gap without collapsing, though at a cost in retention (Appendix G.17). EO shows the same qualitative pattern, in that DER++ has the largest gap (0.325), while FAIR-GEM (0.082), Fair-DER++ (0.086), FSW (0.063), and collapsed GroupDRO (0.025) are substantially lower (Table 2). Fairness forgetting is therefore separable from accuracy forgetting, because methods with similar retention occupy very different fairness points.

D Additional Results

D.1 Readout Choice and Encoder Rankings

Repair recovers accuracy and disparity, which raises the obvious follow-up question of whose encoder was best preserved, which our protocols do not settle. Which encoder appears best preserved depends on the readout used to measure it, so we limit the claims drawn from Sec. 4.2 accordingly.

Table 2: **Our constraints cut DER++'s accuracy-parity gap while keeping most of its accuracy.** Spurious Shift, averaged over tasks. GroupDRO's small gaps come with collapsed accuracy, and its BWT does not measure retention because it resets each task. †: joint training. ⋆: ours. Bold: best continual learner. Mean ± SD over ten seeds.

Method	BalAcc ↑	Acc-BWT ↑	Δ_{AP} ↓	Δ_{EO} ↓	F-BWT$_{\mathrm{AP}}$ ↑	Worst-grp ↑
Joint retrain †	.876±.002	−.016±.002	.271±.006	.281±.007	−.056±.005	.659±.006
Fine-tune	.576±.004	−.416±.006	.239±.007	.250±.008	−.075±.006	.385±.006
A-GEM	.636±.012	−.342±.015	.288±.010	.300±.010	−.108±.008	.407±.009
DER++	**.867±.003**	−.035±.005	.313±.011	.325±.013	−.083±.011	.617±.011
GroupDRO	.532±.006	−.442±.008	**.016±.007**	**.025±.008**	**−.000±.007**	.520±.009
FaIRL-insp.	.672±.019	−.208±.014	.173±.035	.192±.038	−.034±.022	.534±.016
FSW	.779±.012	−.058±.014	.045±.033	.063±.034	−.021±.036	.744±.021
FAIR-GEM ⋆	.741±.015	−.191±.019	.070±.012	.082±.013	−.023±.011	.686±.015
Fair-DER++ ⋆	.837±.004	**−.011±.002**	.073±.011	.086±.012	−.021±.013	**.779±.006**

Two choices define an encoder readout. The *fitting set* is either each method's own retained memory (recoverable in deployment, but differing across methods and so confounding encoder quality with buffer composition) or a held-out pool identical for every method. The *readout* is either a fitted linear head or a training-free assignment to class centroids or nearest neighbours. The disagreement falls entirely on the second axis.

Memory-refit ranks FAIR-GEM first among the continual learners (0.732, against DER++ at 0.700). Both training-free readouts invert that independently and agree with each other on the leading order: kNN puts DER++ first (0.719) and FAIR-GEM fourth (0.683), and NCM does the same (0.699 against 0.611). Neither inversion is seed noise. The centroid rule is moreover the one Rebuffi et al. [2017] classify with, computed here from all training data rather than from exemplars, so a calibrated-head baseline of that family is already among the readouts that disagree.

Fitting one head for every method on the same held-out pool explains why, by finding almost nothing to rank. Seven of nine fall between 0.739 and 0.751, about one point, against the several points that separate the same encoders under kNN or NCM. A fitted head compensates for representational differences that a centroid rule cannot, so the ranking is largely manufactured by the readout. We therefore draw only the claim that survives all four protocols. Every collapsed encoder supports a far better readout than its deployed head.

D.2 Validity of the Loss-Gap Surrogate

Both methods optimise a group-loss gap rather than the reported fairness metric, so the surrogate's validity is still an empirical question. We measure Spearman correlation between the two, within method and pooled, on both benchmarks.

On Spurious Shift the unconditional gap tracks error-based fairness closely (within-method $\rho = 0.910$, $n = 1890$). On Bias in Bios the relationship is less clear: pooled across all methods the correlation with equalized odds is 0.002, though it recovers to 0.695 only when conditioned on the label and restricted to non-collapsed models. Demographic parity is weakly tracked on both ($\rho = 0.113$–0.119 pooled on Spurious Shift), so we make no DP claim for the surrogate. We report within-method correlation alongside pooled because pooled correlation can be produced by between-method spread alone. The full tables, both conditioning variants and both benchmarks, are in Appendix G.28.

E Guarantee Versus Realization

E.1 Scope of the Guarantee

The disparity safety gradient we introduce through FAIR-GEM & Fair-DER++ has both strict guarantees and limitations which we specify here. *Exact:* in exact arithmetic, before downstream clipping and optimizer transformation, Fair-DER++'s single projection is the gradient nearest to $g_{\mathrm{DER++}}$ within its disparity half-space. The implementation's 10^{-8} normalization stabilizer introduces numerical tolerance for nonzero g_{safe}. FAIR-GEM's finite cyclic loop instead targets feasibility

in the intersection and is not claimed to be the nearest dual-constraint solution. *Local:* under a first-order approximation and a sufficiently small step along $-\tilde{g}$, the surrogate change satisfies $D(\theta-\eta\tilde{g})-D(\theta) = -\eta\, g_{\text{safe}}^{\top}\tilde{g}+O(\eta^2) \le -\eta\|g_{\text{safe}}\|_2\varepsilon_t+O(\eta^2)$, where $D = \mathcal{L}_{\text{worst}} - \mathcal{L}_{\text{best}}$ away from max/min ties. *Non-claims:* these properties imply neither convergence to a globally optimal fair predictor nor guaranteed improvement of any non-differentiable fairness metric. Adaptive preconditioning can rotate the applied update relative to the projected gradient. We therefore measure the constraint on norm-matched effective descent directions and test an independently tuned plain-SGD control. Results appear below and in Appendix G.20. The guarantee is also about the disparity an update introduces, which is separate from disparity the data already carries. Joint retraining, having no sequence to forget, allows us to roughly separate the two sources of disparity. A continual learner that exceeds joint retraining in disparity has acquired part of it from the sequence. The comparison does not bound what the constraint can remove, however: on Spurious Shift, Fair-DER++ ends well below joint retraining (0.073 against 0.271). Whether the constraint registers on a benchmark depends instead on the checks in Sec. 4.3.

F Method Definitions

The two disparity-constrained methods appear among the nine compared and are quoted in the synthetic results. They are instruments here rather than the contribution, so their definitions sit outside the body.

Positioning among continual and fairness methods. Catastrophic-forgetting methods fall into regularization [Kirkpatrick et al., 2017, Zenke et al., 2017], replay [Rolnick et al., 2019, Buzzega et al., 2020], and gradient-projection [Lopez-Paz and Ranzato, 2017, Chaudhry et al., 2019, Saha et al., 2021] families, none of which constrains group disparity. Multi-task gradient surgery [Yu et al., 2020, Liu et al., 2021] also edits update directions, but it reconciles objectives the model is meant to satisfy jointly. Ours is asymmetric: accuracy is the objective, and the disparity surrogate enters only as a margin the update must clear. Composing the two is untested here. Static fairness methods [Sagawa et al., 2020, Basu Roy Chowdhury and Chaturvedi, 2023] assume a fixed distribution and do not address sequential drift. At their intersection, Jiang et al. [2023] and Ceccon et al. [2024] document fairness degradation under continual updates without proposing a geometric remedy, and FSW [Park et al., 2026] targets a related gradient-misalignment mechanism through per-epoch sample reweighting rather than constraining the update direction. The fairness-specific hazard (collapsed heads flattening disparity gaps by artifact) and the post-deployment retuning budget are new. We constrain the update itself, and treat accuracy and fairness retention as distinct axes. Full discussion in Appendix G.30.

F.1 Disparity Safety Gradient

The challenge is distinguishing *legitimate* features (those equally informative across all groups) from *spurious shortcuts* that help the majority at the minority's expense, without knowing in advance which features are which. Our approach exploits a geometric property of group-conditional losses.

For group g in the current batch, let $\mathcal{L}_g = \frac{1}{|B_g|}\sum_{i\in B_g}\ell\big(f_\theta(\mathbf{x}_i),\, y_i\big)$ be the mean cross-entropy loss, and let $\mathcal{L}_{\text{worst}} = \max_g \mathcal{L}_g$, $\mathcal{L}_{\text{best}} = \min_g \mathcal{L}_g$. We define the *disparity safety gradient*:

$$g_{\text{safe}} = \nabla_\theta\big[\mathcal{L}_{\text{worst}}(\theta) - \mathcal{L}_{\text{best}}(\theta)\big], \tag{2}$$

computed in a single backward pass. $\mathcal{L}_{\text{worst}} - \mathcal{L}_{\text{best}}$ is a *piecewise-differentiable group-loss disparity surrogate*. We do not equate it with equalized odds or demographic parity, but validate empirically that it tracks prediction-space fairness metrics (Sec. D.2). A direction along which every group's risk changes at the same rate cancels in the local max–min difference, and directions along which the groups disagree survive (formalization in Appendix G.6). The method does not require feature-level annotations of which directions are spurious.

A fixed projection strength risks over-constraining learning, particularly when combined with A-GEM's memory constraint. We therefore use the normalized axis $\hat{g}_{\text{safe}} = g_{\text{safe}}/\|g_{\text{safe}}\|_2$ for nonzero g_{safe} (the implementation adds 10^{-8} to the denominator) and define a dynamic safety margin:

$$\varepsilon_t = \min\big(\alpha,\ \ \beta\cdot(\mathcal{L}_{\text{worst}} - \mathcal{L}_{\text{best}})\big). \tag{3}$$

When the groups perform similarly, $\varepsilon_t \approx 0$ and the constraint loosely prevents any increase in the surrogate. As disparity grows, $\varepsilon_t \to \alpha$, actively requiring a disparity-reducing component in the projected gradient.

F.2 Fair-DER++: Replay Loss + Fairness Projection

DER++ already supplies a loss-level replay signal (Appendix G.8). To isolate the added fairness correction rather than introduce another retention mechanism, Fair-DER++ applies only the fairness projection after the combined DER++ backward pass:

$$\tilde{g} = g_{\mathrm{DER++}} + \max\big(\varepsilon_t - \hat{g}_{\mathrm{safe}}^{\top} g_{\mathrm{DER++}},\ 0\big)\, \hat{g}_{\mathrm{safe}}. \tag{4}$$

For an exactly unit-normalized $\hat{g}_{\mathrm{safe}}$, this is the Euclidean-nearest projection onto one half-space, the same rank-one operation used in FAIR-GEM's alternating loop, but applied once. The safety gradient is computed on the current batch only as memory from past fair tasks would dilute the disparity signal. (Architecture diagram in Appendix G.4)

G Supplementary Material

G.1 Code

Code, configurations, per-seed results, and the scripts that produce every table and figure are available at `https://github.com/MlsBli/broken-model-reads-fairest`.

G.2 Detailed Limitations

Our projection provides a per-step constraint on the linearized group-loss disparity; it does not imply convergence to a globally optimal fair predictor or guaranteed improvement in a non-differentiable fairness metric. Under adaptive preconditioning, the norm-matched effective descent direction satisfies the constraint only partially on diagnostic updates sampled every fourth step (Sec. E.1). The $\max / \min$ surrogate is non-smooth at ties; automatic differentiation follows the currently selected extremal groups. In the recorded diagnostic steps, the worst-group identity switches on average in 27–33% of transitions on Spurious Shift and 40–41% on Bias in Bios. The code exposes EMA ranking as a pilot option, but we have not conducted a controlled multi-seed outcome study of temporal smoothing. Although $\max / \min$ accepts $G > 2$ joint intersectional groups, per-cell estimates become sparse; our experiments use a binary attribute, and the 25-class estimator noise suggests that conditioning and variance control matter. Replay uses capacity 800 (realized 799) per task in the main study; a ten-seed Spurious Shift control covers 200 and 1,600 examples without retuning, but analogous Bias-in-Bios scaling remains untested. The Bias-in-Bios token budget is the reference loader's 32 and discards most of each biography; Appendix G.27 quantifies it and explains why the direction of the readout result should survive a larger budget while its magnitude may not. Partially observed or noisy attributes and larger-scale adaptation remain future work.

G.3 FAIR-GEM Model Architecture

In each step of FAIR-GEM training, we use backpropagation to independently compute three gradients:

g_{task}	Computed from the normal cross-entropy loss of the current batch.
g_{safe}	Computed from the safety disparity as explained in Section F.1.
g_{mem}	Computed from an episodic memory buffer $\mathcal{M}$, which is stratified by sensitive group.

Then, we apply cyclic half-space projections (Appendix G.7) to move g_{task} toward the feasible intersection determined by g_{mem} and the normalized disparity axis $\hat{g}_{\mathrm{safe}}$.

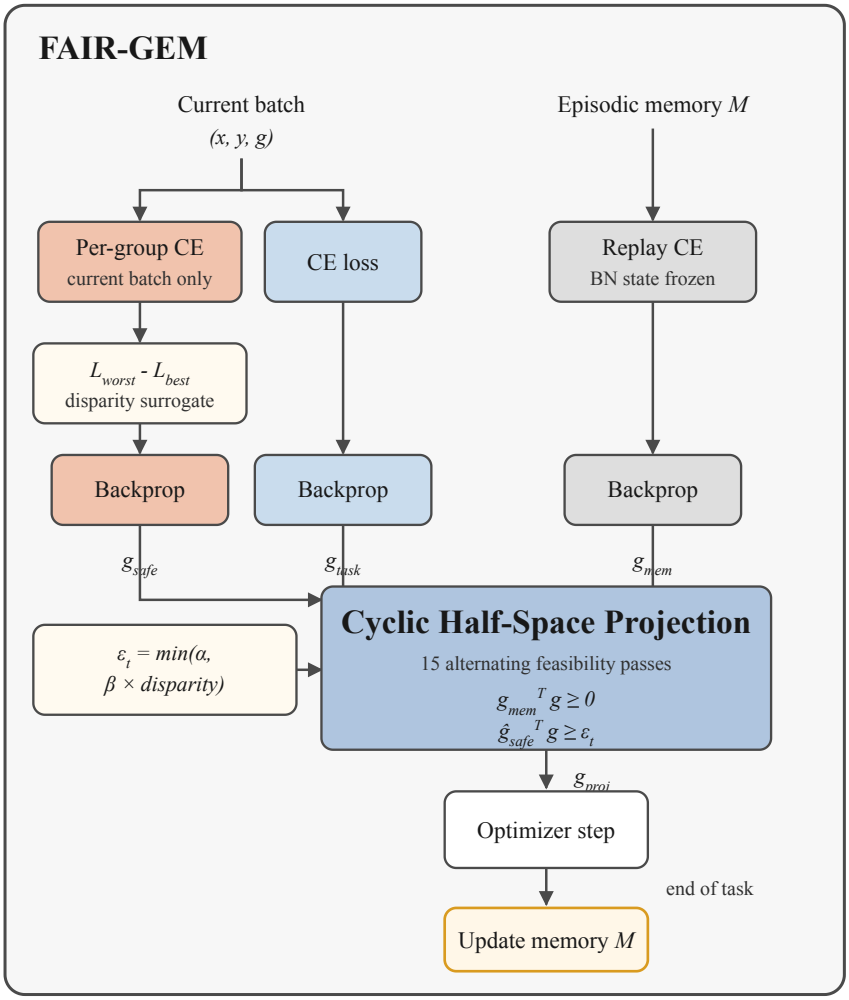


Figure 4: **FAIR-GEM update.** The batch gradient is projected in turn onto a memory-retention half-space and a disparity half-space before the optimizer step.

G.4 Fair-DER++ Model Architecture

In each step of Fair-DER++ training, we use backpropagation to independently compute two gradients:

$g_{\text{DER++}}$	As with the base DER++ model, we augment current-batch cross-entropy with MSE between current and stored logits on one memory draw and cross-entropy against stored hard labels on an independent memory draw.
g_{safe}	Computed from the safety disparity as explained in Section F.1.

Then, we use a single projection (Eq. 4) to map $g_{\text{DER++}}$ onto the half-space whose normal is $\hat{g}_{\text{safe}}$.

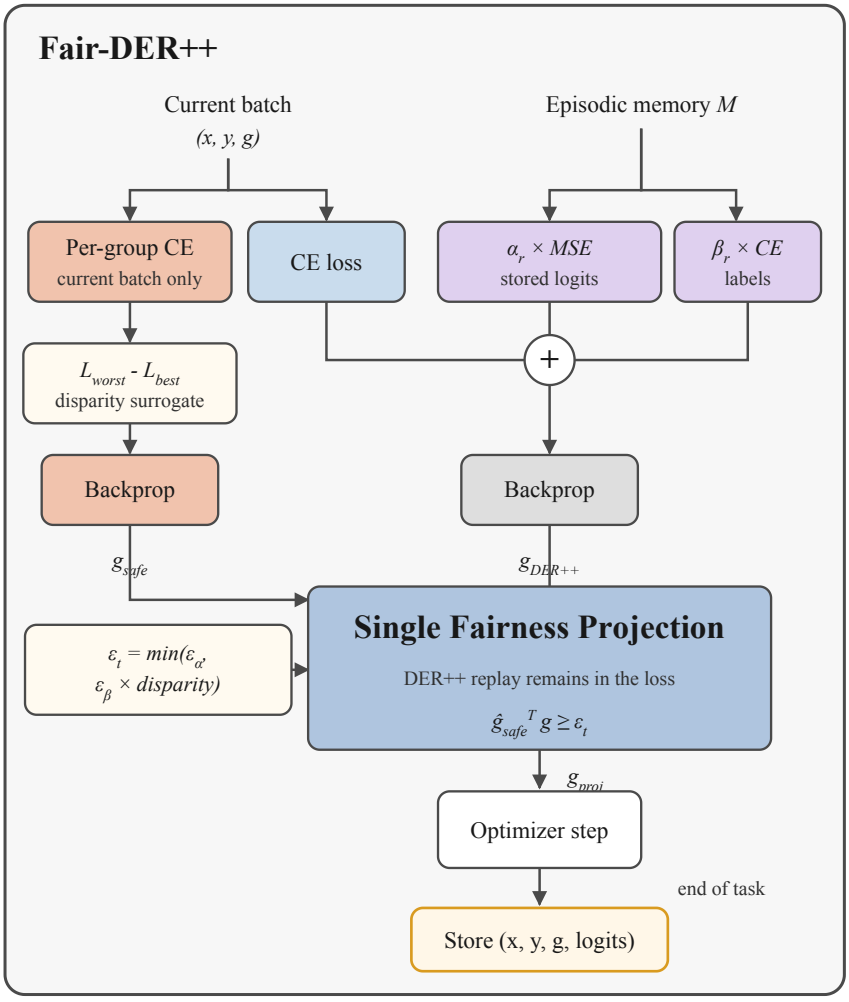


Figure 5: **Fair-DER++ update.** The DER++ gradient is projected once onto the disparity half-space.

G.5 Spurious Shift: Data-Generating Process

For task t, each sample (x, y, g) is built from the label signal $\tilde{y} = 2y - 1$ and group sign $\tilde{s} = 1 - 2g$ (positive for majority group $g = 0$ and negative for minority group $g = 1$):

$$f_{\text{active}} = \ell_t\, \tilde{y} + \epsilon_{\text{a}}, \qquad f_{\text{inactive}} = -0.3\, \tilde{y} + \epsilon_{\text{i}},$$
$$f_2 = c_t\, \tilde{s}\, \tilde{y} + \epsilon_2, \qquad f_j = \epsilon_j \quad (j = 3, \ldots, 19), \tag{5}$$

Each task draws 4,000 training, 2,000 validation and 10,000 test examples, so the 20% minority group contributes roughly 400 validation and 2,000 test samples—comfortably above the 100-sample eligibility threshold the fairness metrics apply (Sec. B.3). The noise terms are $\epsilon_{\text{a}}, \epsilon_{\text{i}}, \epsilon_2, \epsilon_3, \ldots, \epsilon_{19} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and (ℓ_t, c_t) from Table 3. $(f_0, f_1) = (f_{\text{active}}, f_{\text{inactive}})$ on tasks 0–2 and is swapped on tasks 3–5. A single `StandardScaler` fit on task-0 data is applied to all tasks, so later (weaker-ℓ_t) tasks appear proportionally fainter, mildly amplifying the rotation effect. Groups are $0.8/0.2$ (majority/minority) with labels balanced within group, so no base-rate disparity exists and any fairness gap is attributable to the model exploiting f_2. The -0.3 anti-signal is what tests catastrophic forgetting: aggressive re-weighting on tasks 3–5 pushes predictions below random chance on tasks 0–2 rather than just in-line with random chance.

Table 3: **Spurious Shift task schedule.** ℓ_t: label signal in the active feature. c_t: shortcut strength. The task-3 switch from f_0 to f_1 drives accuracy forgetting, and the rising c_t drives fairness forgetting.

Task	ℓ_t	c_t	Active	Dominant challenge
0	2.0	0.0	f_0	Clean baseline
1	1.8	0.5	f_0	Shortcut emerging
2	1.5	1.5	f_0	Gradient conflict
3	1.2	2.5	f_1	Rotation (stale f_0 inverts)
4	1.0	3.0	f_1	Shortcut dominant
5	0.8	4.0	f_1	Max forgetting pressure

G.6 Inter-Group Cancellation Property

Proposition 1 (Cancellation)**.** *Let $\bar{\mathcal{L}}_g(\theta) = \mathbb{E}[\ell(f_\theta(X), Y) \mid G = g]$. Suppose the population worst and best groups are unique and locally unchanged at θ. If $\partial_{\theta_j} \bar{\mathcal{L}}_g(\theta) = c_j$ for every eligible group g, then $\partial_{\theta_j}[\max_g \bar{\mathcal{L}}_g - \min_g \bar{\mathcal{L}}_g] = 0$.*

Proof. Let a and b be the unique local maximizer and minimizer. In their neighborhood, the derivative is $\partial_{\theta_j} \bar{\mathcal{L}}_a - \partial_{\theta_j} \bar{\mathcal{L}}_b = c_j - c_j = 0$. □

Identical feature–label correlation alone is not sufficient for this equality: the groups' conditional feature distributions and current predictions also matter. The result therefore explains cancellation under matched expected gradients rather than identifying every spurious feature in general. The finite-batch estimator used in training need not cancel exactly because of sampling noise, missing groups, or changes in the selected extrema.

G.7 Cyclic Half-Space Projection

FAIR-GEM alternates projections of g_{task} onto $\mathcal{C}_{\text{mem}} = \{\mathbf{g} : g_{\text{mem}}^\top \mathbf{g} \geq 0\}$ and $\mathcal{C}_{\text{fair}} = \{\mathbf{g} : \hat{g}_{\text{safe}}^\top \mathbf{g} \geq \varepsilon_t\}$, targeting their intersection. Each single-half-space projection $\Pi_{\mathcal{C}}$ for $\mathcal{C} = \{\mathbf{g} : \mathbf{a}^\top \mathbf{g} \geq b\}$ is the rank-1 update:

$$\Pi_{\mathcal{C}}(\mathbf{g}) = \mathbf{g} + \frac{\max(b - \mathbf{a}^\top \mathbf{g}, 0)}{\|\mathbf{a}\|^2} \mathbf{a}. \tag{6}$$

Here $g_{\text{safe}} = \nabla(\mathcal{L}_{\text{worst}} - \mathcal{L}_{\text{best}})$ points toward increasing disparity, while the optimizer applies $-\tilde{g}$; hence $\hat{g}_{\text{safe}}^\top \tilde{g} \geq \varepsilon_t$ requires a disparity-reducing component in the parameter update. The released implementation performs 15 cycles, each visiting both half-spaces (at most 30 rank-one projections). If the intersection is nonempty, repeated alternating projections converge to a feasible point, but the fixed 15-cycle implementation has no convergence test. It also contains no Dykstra correction variables, so we claim approximate feasibility behaviour, not exact finite-cycle feasibility or the Euclidean-nearest point in the intersection.

G.8 DER++ Replay Loss

DER++ [Buzzega et al., 2020] adds two replay terms to the current-task loss:

$$\mathcal{L}_{\text{DER++}} = \mathcal{L}_{\text{CE}}(\mathbf{x}_t, y_t) + \alpha_r \left\| f_\theta(\mathbf{x}_m^d) - \mathbf{z}_m^d \right\|_2^2 + \beta_r \mathcal{L}_{\text{CE}}(\mathbf{x}_m^r, y_m^r), \tag{7}$$

where the d and r superscripts denote independent memory draws and $\mathbf{z}_m^d$ are logits stored when the sample was added. The MSE term distills the "dark" logits while the CE term maintains hard labels. On BIOS, stored targets span the fixed 25-way head, and MSE at each later step uses the class positions introduced through that step. Thus newly introduced positions of older stored targets were not previously trained class outputs. Memory forwards use BatchNorm in eval mode so replay does not alter running statistics. Fair-DER++ applies Eq. 4 to $g_{\text{DER++}}$; omission of a separate memory-gradient constraint is a design choice, not a redundancy theorem.

G.9 Stratified Episodic Memory

Uniform memory under-represents the 20% minority group, so we stratify by group. Every group with ≥ 10 samples is first allocated 5 slots. Remaining slots are allocated proportionally to group size, and samples are drawn uniformly without replacement within each group. The configured capacity is 800 samples/task for all replay methods. Because proportional extras are integer-truncated, the reported two-group campaigns retain 799 samples/task; this realized count is equal across replay methods. DER++ and Fair-DER++ also store logits $\mathbf{z}_m$ (at initial insertion time).

G.10 Bias in Bios Preprocessing and Task Construction

We use the Bias in Bios dataset consisting of biography text labelled with profession and gender. We use the `hard_text` field, which has personal identifiers (names and locations) scrubbed, following the original dataset construction. Following prior class-incremental continual-learning protocols, the 25 most frequent professions are sorted by descending training-set frequency and partitioned into five sequential task blocks of five professions each, so that Task 0 contains the five most common professions. Tasks are presented sequentially and models do not revisit previous task data except through episodic replay.

Biographies are tokenized using the BERT tokenizer (`bert-base-uncased`) corresponding to the encoder initialisation, following Basu Roy Chowdhury and Chaturvedi [2023]. Inputs are truncated and padded to a maximum sequence length of 32 tokens, which is the reference loader's setting rather than a tuned choice. This budget is aggressive and we state its effect rather than only its value: over $20{,}000$ sampled training biographies the untruncated length averages 81.1 wordpieces (median 74), so at 32 tokens 97.7% of biographies are truncated and 39.4% of wordpieces are retained; 64 and 128 tokens would retain 70.6% and 96.4% (`analysis/measure_truncation.py`, which needs no training run).

Two consequences, one of which cuts against us. The budget is applied identically to every method including joint retraining, so the comparisons here are internally controlled and the readout-versus-representation result is a within-protocol contrast: the refit reaches 0.732 against joint retraining's 0.748 on the same 32-token inputs, which is direct evidence that the representation supports that accuracy under this budget. But a shorter input yields a less discriminative encoder, which plausibly makes the head more prone to collapsing onto recent classes. The direction of the effect should therefore survive a larger budget while its magnitude may not, and we have not tested that: it requires re-running the campaign at 64 or 128 tokens. Absolute accuracies here are also capped by the budget and should not be read as this encoder's ceiling on the task.

We bounded the part of this that does not need a campaign re-run. Freezing pretrained BERT and fitting a linear probe on its pooled output at each budget, over the same 25 professions, profession balanced accuracy rises from 0.5155 at 32 tokens to 0.5834 at 64 and 0.6010 at 128 — most of the deficit (0.068 of 0.086) is recovered by 64, and the curve is close to flat after that. So the budget does cost real profession information and the cost is bounded rather than open-ended. Gender is different: it is recoverable at 0.9986 AUC or better at every budget, so the protected attribute is already saturated in a 30-wordpiece representation.

That asymmetry is what scopes our claims. The disparity and leakage results rest on gender being present in the representation, which the budget does not affect. The accuracy-level magnitudes — how far the deployed head falls, how much the refit recovers — are the ones a larger budget could move. This probe is a bound on input information, not a substitute for the experiment: it uses the frozen pretrained encoder rather than the continually fine-tuned one, a single split with no seed replication, and it says nothing about collapse dynamics. Reproduced by `analysis/probe_token_budget.py`, which needs no GPU. Within each task, 15% of training examples are held out as a validation split, stratified jointly by profession and gender to ensure all demographic–class combinations are represented. Gender serves as the sensitive attribute for all fairness evaluations.

For deployed-head evaluation, the classifier is trained incrementally using the corresponding continual-learning algorithm. For encoder-level evaluation, we freeze the trained encoder after the final task and refit a class-balanced multinomial logistic-regression head on each replay method's final 3,995-example memory. For methods without memory, the released diagnostic uses a seeded 4,000-example nominal-budget sample from all observed training data. This removes deployed-head recency but

remains confounded by the retained fitting data; it is not the deployed classifier or a shared-data encoder ranking.

The refit head is scikit-learn's `LogisticRegression` at its default inverse-regularization strength $C = 1$ with the lbfgs solver, identical for every method and every budget. We did not tune it. Every refit number reported here is therefore a measurement at one regularization setting rather than the best achievable refit, which matters most for the budget–recovery crossover: where a non-collapsed model's refit overtakes its deployed head is a property of this fixed head, and a more strongly regularized or calibrated head would plausibly move that point down. The comparison across methods is unaffected, because all of them are refit identically.

G.11 BertMLP Architecture (BIOS)

Encoder: `bert-base-uncased` (12 layers, 768-d, $\sim$110M params) with gradient checkpointing and an attention mask from non-pad tokens.
Head: $768 \to 256$ (BN, ReLU, dropout 0.1) $\to 256$ (BN, ReLU, dropout 0.1) $\to 25$.
Logit masking: unseen-class logits are set to $-\infty$ before softmax in both training and evaluation. This prevents future, unintroduced classes from entering the softmax denominator; newly introduced classes compete normally with old classes.
Optimizer: AdamW with two parameter groups—BERT at $\sim 2\times10^{-5}$ (per-method, tuned), head at 3×10^{-4}, weight decay 10^{-4}. Gradient projection (A-GEM, FAIR-GEM) operates on the full $\sim$110M-parameter vector, constraining BERT updates and not just the head.

G.12 FAIR-GEM Training Algorithm

Algorithm 1 FAIR-GEM training step on task t

Require: batch B, memory $\mathcal{M}$, margin params α, β
$g_{\text{task}} \leftarrow \nabla_\theta \mathcal{L}_{\text{CE}}(B)$
if $\mathcal{M} \neq \emptyset$ **then**
  $B_m \sim \mathcal{M}$ (up to 448 samples); $g_{\text{mem}} \leftarrow \nabla_\theta \mathcal{L}_{\text{CE}}(B_m)$ {BN eval mode}
end if
if B contains at least two eligible groups **then**
  $g_{\text{safe}} \leftarrow \nabla_\theta[\mathcal{L}_{\text{worst}} - \mathcal{L}_{\text{best}}]$; {current batch only}
  if $\|g_{\text{safe}}\|_2 > 0$ **then**
    $\hat{g}_{\text{safe}} \leftarrow g_{\text{safe}}/\|g_{\text{safe}}\|_2$; $\varepsilon_t \leftarrow \min(\alpha,\ \beta(\mathcal{L}_{\text{worst}} - \mathcal{L}_{\text{best}}))$
  end if
end if
Spurious Shift only: clip $\|g_{\text{task}}\|_2 \leq 3.0$
$\tilde{\mathbf{g}} \leftarrow$ AlternatingProject(g_{task}; available constraints, 15)
Bias in Bios only: clip $\|\tilde{\mathbf{g}}\|_2 \leq 3.0$
$\theta \leftarrow$ AdamWStep($\theta, \tilde{\mathbf{g}}$)
after task: $\mathcal{M} \leftarrow \mathcal{M} \cup$ StratifiedSample(t, capacity 800)

G.13 Additional Spurious Shift Figures

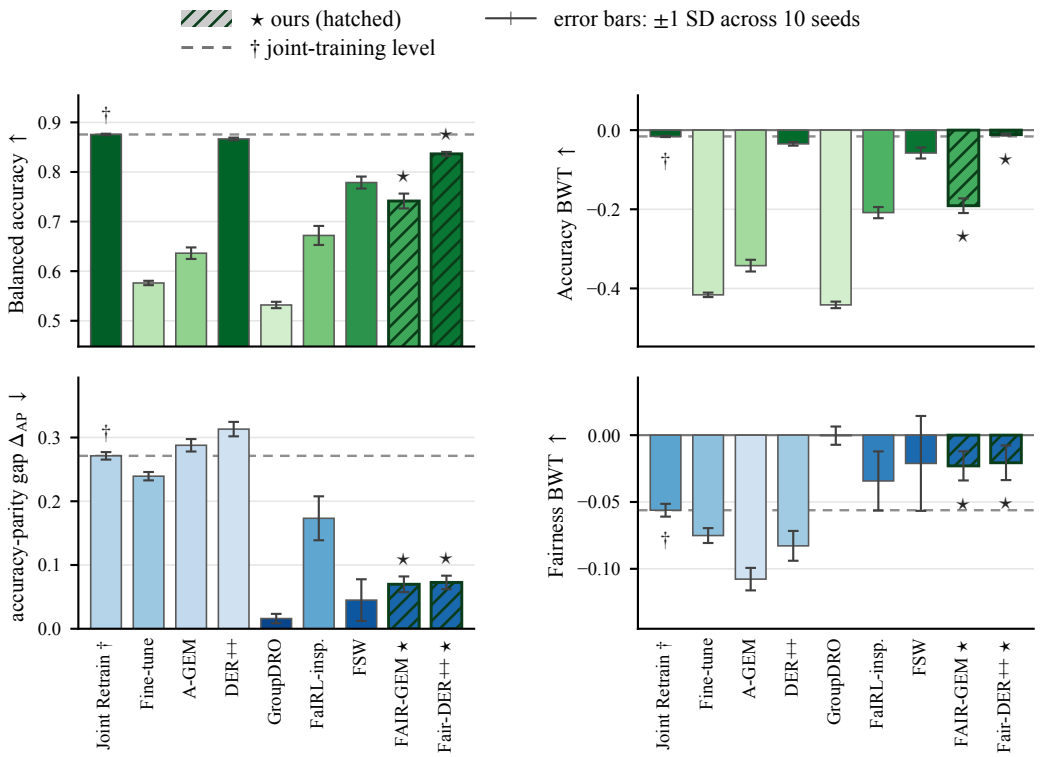


Figure 6: **DER++ keeps its accuracy on Spurious Shift but has the largest fairness gap, and our constraints cut it.** Final accuracy and accuracy BWT (top), accuracy-parity gap and fairness BWT (bottom). Hatched: ours. Dashed: joint training. Error bars: ±1 SD over ten seeds.

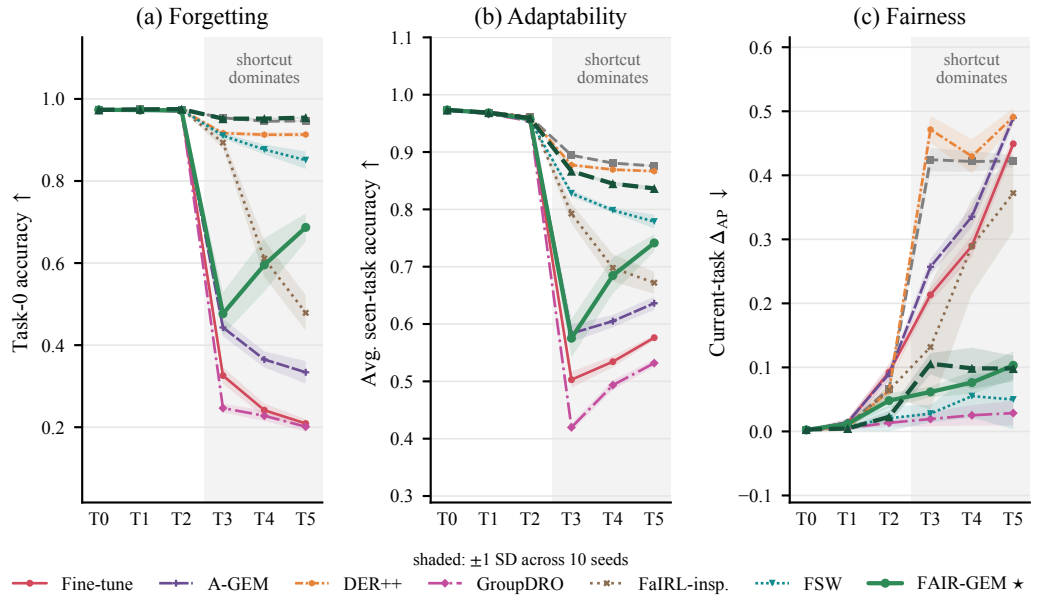


Figure 7: **Accuracy and fairness forget separately on Spurious Shift.** (a) Task-0 accuracy. (b) Mean accuracy over seen tasks. (c) Current-task accuracy-parity gap, which grows once the shortcut dominates (grey). Shading: ±1 SD over ten seeds.

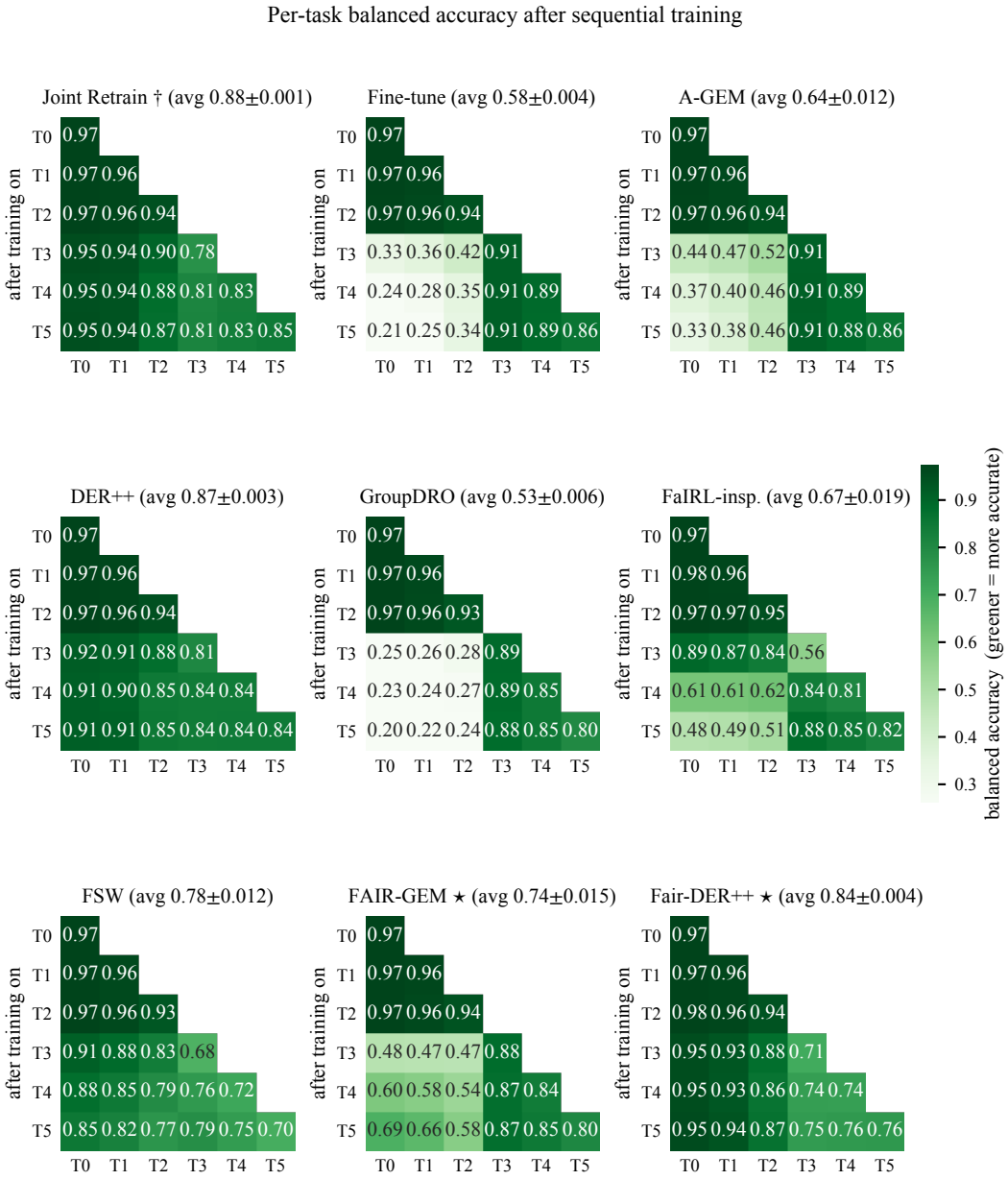


Figure 8: **Most forgetting on Spurious Shift happens at the task-3 feature rotation.** Accuracy on task i (column) after training through task j (row), averaged over ten seeds.

G.14 Ablation Studies (Spurious Shift)

We ablate the gradient direction, margin schedule, memory sampling, and margin sensitivity on Spurious Shift, retraining each variant over ten seeds while holding the remaining method fixed. **Static** ε replaces the dynamic margin (Eq. 3) with a fixed value equal to the mean ε_t the dynamic schedule actually applies, matching average constraint strength. **Avg grad** replaces the worst−best disparity gradient with the average-group gradient. **Top-k DRO** replaces it with a top-k loss on the worst group (FAIR-GEM only). **Uniform memory** removes group stratification. Finally, α and β (or $\varepsilon_\alpha, \varepsilon_\beta$ for Fair-DER++) are varied by $\pm 25\%$. The direction ablations are decisive: Avg grad raises AP from 0.070 to 0.165 for FAIR-GEM and from 0.073 to 0.183 for Fair-DER++, while top-k DRO reaches 0.160. Static ε has different backbone-specific effects: FAIR-GEM loses accuracy ($0.741 \rightarrow 0.702$) at similar AP, whereas Fair-DER++ keeps accuracy but widens AP ($0.073 \rightarrow 0.075$). The $\pm 25\%$ arms and uniform-memory arms remain near the full methods, showing local margin robustness and little dependence on buffer stratification. The wider $k \in \{0, .25, .5, 1, 2, 4\}$ paired sweep in Figure 21 separately tests the global accuracy–fairness response rather than local robustness. A further arm replaces the projection with a loss-level penalty on the same disparity signal, which is the most direct alternative to the mechanism this paper studies; it is reported separately in Appendix G.17 because its verdict is mixed rather than confirmatory.

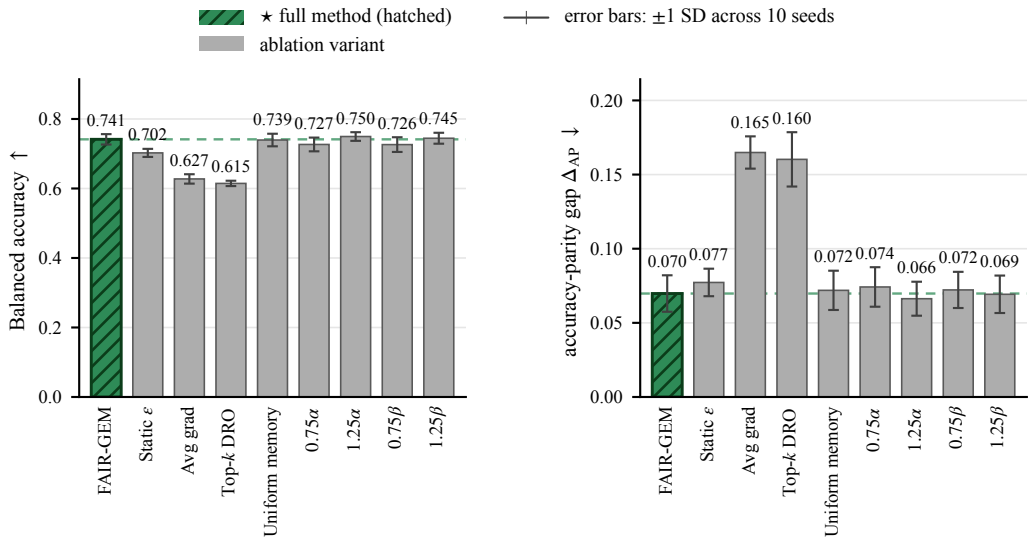


Figure 9: **Changing FAIR-GEM's constraint direction more than doubles its gap, while its margin settings barely matter.** Final accuracy (left) and accuracy-parity gap (right) on Spurious Shift for the full method (hatched) and each ablation. Error bars: ± 1 SD over ten seeds.

G.15 Omissions in Our Environment Record

The recommendation this finding implies is to report the environment, so it is worth saying where our own record stops. We pinned and logged Python, SciPy and the full dependency set, and we report both wall-clock figures and the ratio between them. We did not record the BLAS implementation, the thread count, or the OS build — which, for a difference that lives entirely in a linear-program solve, are the three most likely places for it to come from. Anyone trying to reproduce the $\sim$122 spread should record those first, and we would if we ran it again.

G.16 Multiplicity Across the Permutation Family

Every comparison we test uses the same exact two-sided paired permutation procedure, and there are 17 of them: 9 on Spurious Shift and 8 on Bias in Bios, four metrics for each of our two methods against its own backbone, plus the FAIR-GEM-SGD versus FSW comparison. An earlier version of this work corrected for none of them. A p-value of 0.0020 describes the chance of a difference this

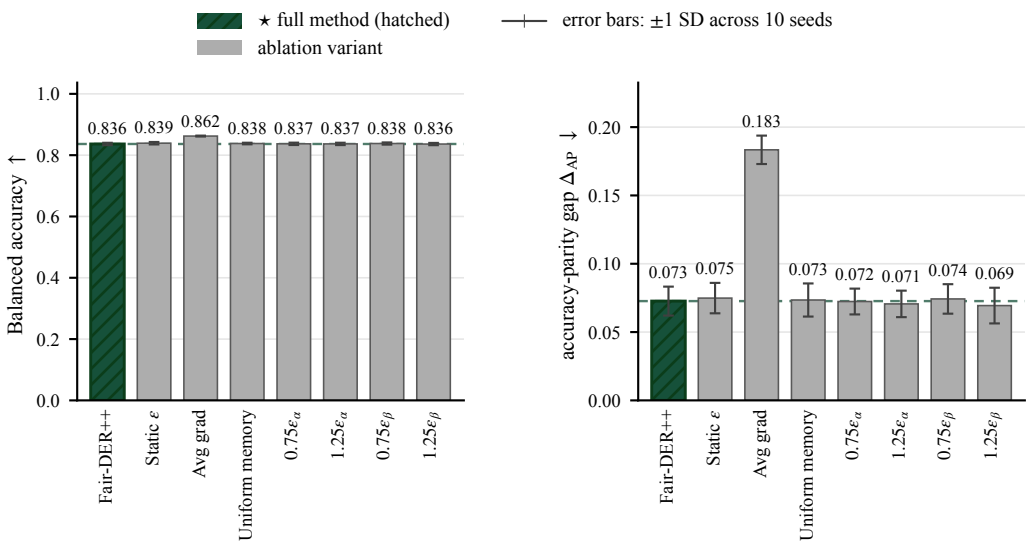


Figure 10: **Only the averaged-gradient direction hurts Fair-DER++, more than doubling its gap.** Layout as in Figure 9.

large under the null for the test performed, and once several tests are performed the relevant question is the chance that the most extreme of them looks this large, which is a different and larger number.

We use Holm–Bonferroni. It controls the family-wise error rate, is uniformly at least as powerful as plain Bonferroni, and assumes nothing about the dependence structure — which matters here, because balanced accuracy, worst-group accuracy, accuracy parity and equalized odds are computed from the same predictions on the same seeds and are strongly dependent. Under dependence Holm is conservative, so a result that survives it survives.

The natural family is one benchmark: a single set of hypotheses about the same methods on the same task sequence. We report the pooled family over both benchmarks as well, because a paper arguing that protocol choices decide results should not be choosing the partition that suits it. Both columns are in `posthoc_studies.csv`.

	Tests	Clearing $p < 0.05$	
Benchmark	m	raw	Holm (per benchmark)
Spurious Shift	9	8	8
Bias in Bios	8	1	0

On Spurious Shift the conclusion is unchanged: all 8 backbone comparisons sit at the floor 0.0020 and clear the threshold after correction at 0.018 per benchmark, or 0.034 pooled. On Bias in Bios it changes: exactly one of the 8 tests reached $p < 0.05$ raw — Fair-DER++ against DER++ on equalized odds, at 0.010, favouring the backbone — and after correction it is at 0.078 per benchmark and 0.088 pooled. Nothing on that benchmark clears the threshold once the family is accounted for. We had described that comparison as significant. It is better described as a difference the ten seeds are consistent about and the design cannot certify at this family size. The claim the correction removed was one we had made against our own method.

The corrected floor. The floor arithmetic extends, and the extension is more useful than the correction. The smallest p-value an exact paired test can return is $2/2^n$; the smallest a Holm-corrected exact paired test can return is $m \cdot 2/2^n$, since the smallest adjusted value in a family of m is the smallest raw value multiplied by m. The seed budget a design needs is therefore a function of how many comparisons it intends to make, not a constant. For the family sizes here:

	Smallest attainable corrected p		
Family	$n = 3$	$n = 5$	$n = 10$
Spurious Shift ($m = 9$)	1.00	0.56	0.018
Bias in Bios ($m = 8$)	1.00	0.50	0.016

At three seeds no corrected result is available at any effect size, and at five the floor is above $p < 0.5$, so the conventional budget in this literature cannot support a multiplicity-corrected claim about a family of this size at all. At ten seeds the floor is 0.018, which clears 0.05 but not by much: a family twice this size would not. The procedural remedy is one line long — report the seed count and the number of comparisons beside the p-value — and both numbers are free.

Residual limitations. A family-wise correction addresses tests we ran and reported. It says nothing about comparisons examined and not reported, and nothing about the choice of which comparisons to run, which was made after seeing the benchmark. We predeclared no analysis plan; the family above is the set of tests the manuscripts contain, and a reader has to take our word that it is also the set we performed.

G.17 Loss Penalty vs. Gradient Projection

The most direct alternative to constraining the update is to constrain the objective: add the same group-loss gap to the loss and let the optimizer trade it off. We ran that arm. **DER++-Penalty** keeps DER++'s replay and distillation terms and its memory budget unchanged and adds $\lambda \cdot \Delta_{\text{loss}}$ to the per-batch objective, with Δ_{loss} the same worst−best group loss gap the projections use as their constraint direction, over the same ten seeds.

It reaches a lower disparity gap than either projection — AP $= 0.017 \pm 0.009$ against 0.070 for FAIR-GEM and 0.073 for Fair-DER++ — and the best fairness backward transfer of any arm we ran (F-BWT $= 0.005 \pm 0.011$, against -0.021 for Fair-DER++). We report this because it is the obvious alternative to our mechanism and it wins on the disparity axis. It also settles a question the main text leaves open: GroupDRO reaches a comparable gap (0.016) only by collapsing to 0.532 balanced accuracy, but that is a fact about GroupDRO rather than a general price of low disparity — a loss penalty gets there at 0.795.

The penalty costs retention, which is the axis a continual learner is judged on. The penalty reshapes the loss surface, and its backbone's accuracy backward transfer degrades from -0.035 to -0.055, while the same disparity signal applied as a projection improves it to -0.011 — the best Acc-BWT of any method we compare, better even than the joint-training reference's -0.016. Final accuracy falls the same way (0.795 against 0.837). The two arms trade rather than order: each buys one axis with the other.

Two reasons we keep the projection as the object of study, neither of which is that it wins this comparison. First, it leaves the objective alone. It needs only a disparity gradient, so it composes with training objectives that cannot be edited — a third-party or frozen loss, or a replay objective already carrying its own distillation terms — and it is a rank-one operation whose cost is fixed per step regardless of model size, which is what lets the same constraint run unchanged at BERT scale (Appendix G.25). A penalty weight has to be inserted into the loss and re-selected per benchmark, and Sec. D.2 is the reason that matters here: where the loss-gap surrogate stops tracking the reported metric, a penalty on it is optimizing the wrong quantity with no margin to check, whereas the projection's realized constraint is directly measurable (Sec. E.1). Second, the margin ε_t is a per-step quantity that adapts to the gap actually observed (Eq. 3); λ is one global constant for the whole sequence.

One caveat, without which the comparison is misleading. This arm ran at its default λ: the equal-budget validation search (Appendix G.26) was configured for it but not completed, so unlike every method in Table 2 it was not tuned under the common objective. That makes it an ablation rather than a protocol-matched entry, and it makes the numbers above a lower bound on what a loss penalty achieves here — tuning λ would be expected to improve it, not to close the gap. A protocol-matched comparison, and the same arm on Bias in Bios where the projections do not transfer, are both open.

G.18 Uniform vs. Group-Stratified Memory

Because our episodic memory stratifies by group (Appendix G.9), a natural concern is that the fairness gains come from the buffer composition rather than the projection. Re-running FAIR-GEM and Fair-DER++ with uniform (unstratified) sampling at the same budget yields FAIR-GEM 0.739 ± 0.018 BalAcc/0.072 ± 0.013 AP and Fair-DER++ 0.838 ± 0.003/0.074 ± 0.012, overlapping the full-method seed variation.

G.19 Normalization Study

Because BatchNorm can couple examples across groups, we separately compare four normalization protocols, retuning every tunable method within each condition and evaluating ten seeds. The safety methods remain effective under LayerNorm, which has no batch coupling or running statistics: FAIR-GEM obtains 0.696 ± 0.057 BalAcc/0.054 ± 0.012 AP and Fair-DER++ 0.850 ± 0.002/0.091 ± 0.010. Performance varies substantially across BatchNorm handling—FAIR-GEM's gap spans 0.054–0.139 and Fair-DER++'s 0.060–0.193 across the four conditions—so normalization is a genuine protocol choice rather than a hidden invariant. Each condition is retuned independently and the study is not pooled with the main campaign, though it now runs at the same ten-seed budget.

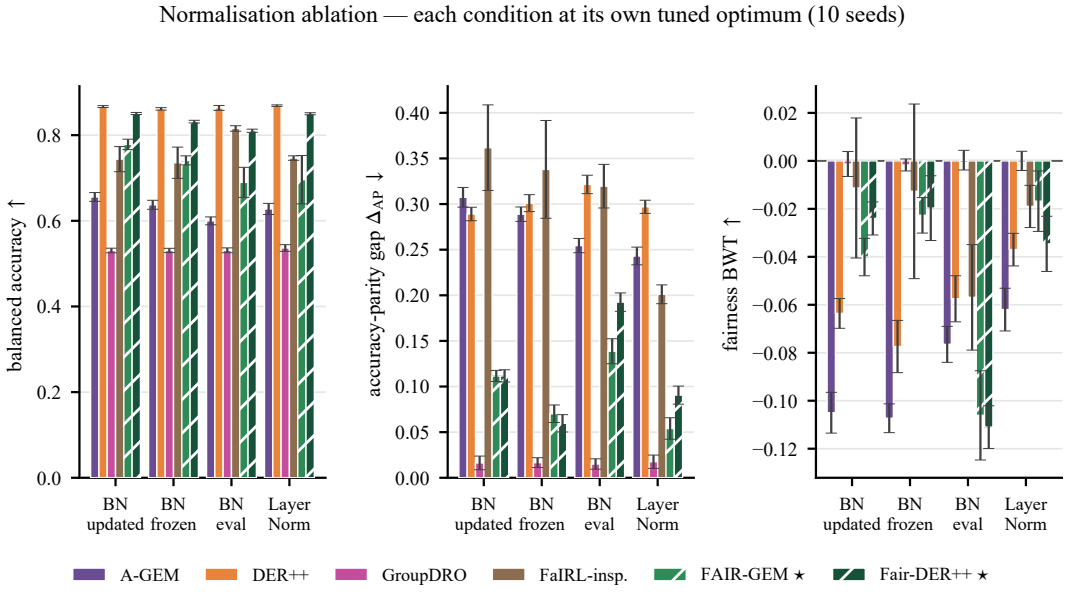


Figure 11: **Our methods do not need BatchNorm, but their fairness depends on how auxiliary passes treat normalization.** Spurious Shift, with each scheme tuned separately. Error bars: ± 1 SD over ten seeds.

G.20 SGD-Projected Variants: Realized-Step Alignment

Under AdamW the norm-matched effective descent direction satisfies the projected disparity half-space on only a minority of sampled steps (Sec. E.1). With plain SGD (no momentum or weight decay), $-\Delta\theta$ is parallel to $\tilde{g}$, making the disparity-half-space condition hold whenever it holds for $\tilde{g}$, up to numerical tolerance. With learning rates tuned independently for the SGD arms (3.0×10^{-3} for FAIR-GEM, 2.3×10^{-2} for Fair-DER++), Spurious Shift results over this arm's own ten-seed budget are: FAIR-GEM-SGD BalAcc 0.747 ± 0.006, $\Delta_{\mathrm{AP}} = 0.033 \pm 0.011$ (vs. 0.741/0.070 under AdamW); Fair-DER++-SGD 0.825 ± 0.003, 0.070 ± 0.012 (vs. 0.837/0.073). The low-disparity FAIR-GEM behaviour therefore persists without adaptive preconditioning; Fair-DER++ changes little in gap while losing some accuracy. FAIR-GEM-SGD's AP gap is numerically close to FSW's 0.045 ± 0.033: the observed difference is 0.012 and an exact paired permutation test over the same ten seeds gives $p = 0.352$, so the two are indistinguishable at this budget rather than merely close.

G.21 Projection-Cycle and Buffer-Size Sensitivity

FAIR-GEM's alternating projection defaults to 15 cycles. Sweeping $\{1, 3, 5, 15, 30\}$ over this arm's own ten seeds, a single cycle gives low accuracy (0.535 ± 0.007), while 3–30 cycles yield similar observed means with overlapping seed variation (BalAcc 0.717–0.736, $\Delta_{\mathrm{AP}} = 0.067$–$0.073$); the 15-cycle default (0.741 ± 0.015, $\Delta_{\mathrm{AP}} = 0.070 \pm 0.012$) lies within this range. Buffer-size controls at 200 and 1,600 samples/task run the same ten seeds as the main campaign, without retuning; the 800-sample point is the ten-seed main result. Budgets are equal across methods within each setting. The comparison shows the fairness effect is budget-robust: Fair-DER++ $\Delta_{\mathrm{AP}} = 0.081/0.073/0.070$ and FAIR-GEM $0.071/0.070/0.073$ across the three budgets, while DER++ remains near 0.313 at every size. These controls show that the observed fairness difference is not specific to the 800-example budget (`sensitivity_summary.csv`).

G.22 Determinism and Cross-Platform Reproduction

We separate two questions that reproducibility claims often conflate.

Determinism. The pipeline is deterministic on a fixed machine. Two runs of the same seed on the same machine under the pinned dependency set produce bitwise-identical result matrices (maximum absolute difference 0 across every metric and method we checked). Every source of randomness—model initialisation, batch order, memory sampling, and the generated dataset—is seeded, so a third party on comparable hardware reproduces our numbers exactly rather than approximately.

Cross-platform reproduction. Across platforms the campaign does not reproduce bitwise, which is expected. Rerunning it under the same pinned versions on different hardware moved the largest single matrix cell by 0.12 balanced accuracy. Cells moved for every method, including those with no fairness machinery, which locates the cause in platform-level floating-point and BLAS behaviour rather than in any method or in our seeding. PyTorch does not guarantee bitwise reproducibility across platforms, and we therefore report the platform (Appendix G.26) rather than claiming hardware-independent determinism.

Conclusion stability across platforms. The reported conclusions do survive the platform change. The aggregates are stable, in that every reported mean shifts by at most 0.004 except FSW, whose accuracy-parity gap moves by 0.015 with roughly doubled seed variation—consistent with the sensitivity of its per-epoch linear program and with the non-monotone sweep response in Figure 21. Every qualitative conclusion replicates: DER++ retains the most accuracy and carries the largest gap, both projection methods reduce their backbones' gaps by roughly a factor of four, and Fair-DER++ records the best retention. Per-method figures are in `replication_summary.csv`. We regard a result that survives a platform change as better evidence than one that reproduces only bitwise on the machine that produced it.

G.23 Provenance of the Realized-Step Diagnostic

A per-step mechanism claim should name the runs it was measured on, so we state it precisely. The realized-step rates in Section E.1 come from the same 10 seeds that produced Table 2, re-executed on the same machine with the step diagnostic enabled. The instrumentation only measures around the optimizer step, and we verified rather than assumed that this leaves training untouched: the instrumented run reproduces the campaign's result matrices exactly, with a maximum absolute difference of zero across every seed, metric and method. The diagnostics therefore describe the runs the tables report, not a parallel set of runs under the same protocol.

The Bias-in-Bios diagnostic is narrower and we state that too. The step diagnostic is opt-in, and the instrumented pass was run for three of the 10 seeds the Bias-in-Bios tables now report, so Figure 17 describes a subset of those runs rather than all of them. It supports the per-step mechanism claim on the runs it covers and we draw no cross-seed distribution from it; extending it is an instrumented re-run of the remaining seeds, not a new measurement.

G.24 Frozen-Encoder Control

To test whether the readout/representation separation depends on encoder plasticity, we rerun the sequence with BERT frozen, each method training only an MLP head on cached embeddings. Absolute levels fall, as a weaker representation implies: joint retraining reaches 0.666 on the deployed head against 0.740 when the encoder is fine-tuned.

Two things carry over. Collapse still occurs—FAIR-GEM's deployed head sits at 0.184, at the same recency ceiling—so a readout failure of this kind does not require the encoder to have drifted during training. And the refit still recovers it, to 0.603, reproducing the main arm's structure at a lower ceiling.

One thing does not carry over: the ranking. Under memory-refit the fine-tuned arm places FAIR-GEM first among the continual learners (0.732), while the frozen arm places it behind joint retraining (0.635), DER++ (0.619) and Fair-DER++ (0.617). One protocol, one dataset, and the same methods ordered differently according only to how much encoder plasticity the setup permits. This is a second axis of the protocol sensitivity the main text reports along the readout axis, and it points the same way: an encoder ranking is a property of the measurement setup as much as of the encoders.

Encoder plasticity and readout disagreement. This arm's nearest-class-mean column initially read exactly $1/25$ for every method, joint retraining included, which does not collapse on the deployed head. The cause was a data-loading defect specific to the frozen configuration: the NCM script requested tasks without frozen encoder embeddings, which is correct while the encoder is fine-tuned—the model then consumes token ids—but with a frozen encoder those embeddings are the model's input, so every method received the same uninformative features and the nearest-centroid assignment degenerated onto the final task's classes. Delegating that flag to the task factory fixes it. The defect could not affect the fine-tuned results, where the corrected call resolves to the previous behaviour, and we re-ran the frozen arm's NCM over all 10 seeds after fixing it.

The corrected column is ordered as one would expect—joint retraining first at 0.702, then Fair-DER++ (0.641) and DER++ (0.640), with plain fine-tuning last among the learners at 0.465—and it supports a claim the main text could not make. In the fine-tuned arm the two representation readouts show no detectable agreement on how to order the methods: Spearman $\rho = 0.267$ across 9 methods ($p = 0.4933$, exact two-sided permutation over rank orderings). With the encoder frozen they agree closely, $\rho = 0.917$ ($p = 0.0013$). Two readings of that contrast are available and we take the weaker one. A non-significant rank correlation over 9 methods is not evidence that the readouts disagree; it is a failure to establish that they agree, and at this number of methods the test cannot separate genuine reordering from insufficient power. We therefore rest the claim on the reordering we observe directly rather than on the correlation contrast: the inversion the main text reports occurs in the fine-tuned arm and is absent in the frozen one. On that evidence the readout disagreement is not an irreducible property of probing representations but is associated with end-to-end plasticity. When every method shares a fixed encoder and only a shallow trunk adapts, memory-refit and NCM measure nearly the same thing. This localizes the protocol sensitivity rather than merely documenting it, and it is why we report which readout a number came from wherever we report one.

G.25 Runtime Overhead, and Its Environment Sensitivity

Relative to their backbones, the projections cost $1.75\times$ (FAIR-GEM vs. A-GEM) and $1.44\times$ (Fair-DER++ vs. DER++) on Spurious Shift, and $1.95\times$/$1.45\times$ on Bias in Bios. This overhead is a fixed per-step cost—one extra backward pass and a rank-one update—so it is stable across environments and does not scale with dataset size.

FSW's cost is not stable, and the instability is worth reporting because it is easy to mistake for a property of the method. Its per-epoch linear program over all current-task samples dominates its runtime, and the solve time depends on which linear-programming backend SciPy resolves `method="highs"` to. We measured the identical configuration—30 epochs per task, 150 solves over $4{,}000$ samples, the same pinned dependency versions and the same tuned hyperparameters—in two environments and obtained $96\times$ DER++ in one and $0.79\times$ in the other, a factor of $\sim$122. The two runs produce statistically indistinguishable results and identical sample-retention behaviour, so the difference is solve time alone. Every other method moved by less than $1.1\times$ between the same two environments, which isolates the cause to FSW's LP call rather than to hardware speed.

Two disclosures this comparison needs and did not carry. The two sides do not share a seed basis: the reference environment is a preserved snapshot taken when the campaign ran five seeds, while the comparison environment is the current ten-seed campaign. Each side is normalised by its own DER++ time, so machine speed divides out and the mismatch adds noise to one side rather than bias to the ratio — but a seed basis belongs next to the claim it supports, and re-measuring the reference side at ten seeds would mean reconstructing an environment we no longer have, which is the same reason the snapshot exists at all. And it varies whole environments rather than one factor at a time: we did not hold hardware, BLAS build, thread count and SciPy version fixed while flipping the solver backend, so we can attribute the spread to the LP call but not to any single component of it. A factor-by-factor sweep is the experiment that would, and we have not run it.

We therefore make no portable claim about FSW's absolute cost, and we withdraw the epoch-budget explanation an earlier version of this analysis offered: the apparent contrast between the two benchmarks confounded epoch count with solver backend, and we cannot separate them from the runs we have. The data supports a narrower claim, and one more useful for a deployment decision: an LP-based intervention inherits the variance of its solver stack, where a fixed-cost projection does not.

On Spurious Shift, FSW retains 36.7% of current-task samples on average; seed–task means range from 28.6% to 60.1%, and this figure is unchanged across both environments. Full per-method wall-clock and sample-retention tables ship with the code.

G.26 Full Hyperparameter Settings

Table 4: **Method-specific hyperparameters.** Tuned with Eq. 1, with 20 trials on Bias in Bios and 15 on Spurious Shift. BERT learning rates are in Table 5.

Parameter	Spurious Shift	BIOS
Shared training		
Head LR / Weight decay	3×10^{-4} / 10^{-4}	3×10^{-4} / 10^{-4}
BERT LR	—	$\sim2\times10^{-5}$ (per-method)
Batch size / Epochs per task	512 / 30 (early stop)	256 / 2
Warmup / Grad-clip norm	3 ep / 3.0	6% steps / 3.0
Memory per task / A-GEM batch	800 / 448	800 / 448
GroupDRO		
η / C	0.0056 / 2.8521	0.0212 / 2.0997
FaIRL-inspired diagnostic		
λ (adversarial) / ε (MCR^2)	4.1749 / 1.3385	0.1661 / 1.9891
DER++		
α_r / β_r	0.1724 / 0.1732	0.3071 / 1.7255
FSW		
α / λ / τ	0.00108 / 0.2866 / 1.4417	5.25×10^{-4} / 0.1531 / 4.3797
FAIR-GEM		
α / β / alternating passes	0.3794 / 3.1940 / 15	0.2444 / 1.8105 / 15
SGD-arm LR (Appendix G.20)	3.0×10^{-3}	—
Fair-DER++		
α_r / β_r (replay)	1.0460 / 0.2601	0.2801 / 1.9243
ε_α / ε_β	0.3753 / 1.1352	0.3792 / 1.0569
SGD-arm LR (Appendix G.20)	2.3×10^{-2}	—

Because the BERT learning rate is tuned per method, the single value quoted in Table 4 is a summary; the selected rates span a factor of 9 and are listed individually below.

Table 5: **Per-method BERT learning rates on Bias in Bios.** Each comes from its own 20-trial Optuna study. The head learning rate is 3×10^{-4} throughout.

Method	BERT LR	Method	BERT LR
Joint retrain	3.67×10^{-5}	FaIRL-insp.	1.66×10^{-5}
Fine-tune	7.16×10^{-6}	FSW	3.79×10^{-5}
A-GEM	4.55×10^{-5}	FAIR-GEM	2.05×10^{-5}
DER++	2.70×10^{-5}	Fair-DER++	5.08×10^{-6}
GroupDRO	5.06×10^{-6}		

G.27 Bias in Bios: Secondary Metrics and Figures

Table 6: **Refit-head fairness gaps are similar across methods on Bias in Bios.** TPR and accuracy-parity gaps on the memory-refit head, and nearest-class-mean (NCM) accuracy. Mean ± SD over ten seeds.

Method	$\Delta_{\mathrm{TPR,mem}}\downarrow$	NCM↑	$\Delta_{\mathrm{AP,mem}}\downarrow$
Joint retrain †	.160±.008	.755±.003	.036±.004
Fine-tune	.163±.011	.595±.017	.044±.006
A-GEM	.173±.025	.585±.021	.037±.005
DER++	.163±.009	.699±.007	.045±.004
GroupDRO	.164±.013	.597±.017	.044±.006
FaIRL-insp.	.148±.017	.534±.014	.041±.005
FSW	.157±.010	.678±.008	.047±.008
FAIR-GEM ⋆	.172±.014	.611±.025	.037±.003
Fair-DER++ ⋆	.165±.009	.664±.004	.043±.005

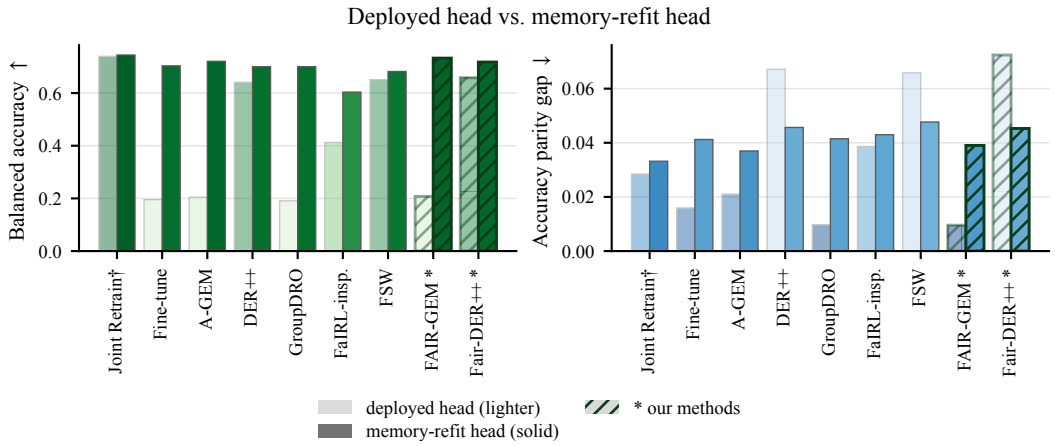


Figure 12: **Refitting the head on retained memory recovers every collapsed method.** Balanced accuracy (left) and accuracy-parity gap (right) for the deployed head (light) and the memory-refit head (solid).

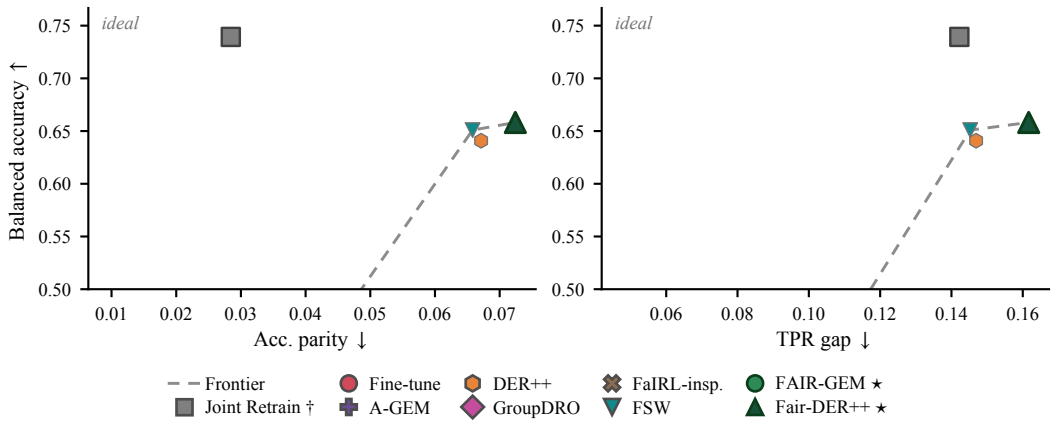


Figure 13: **Among working models, DER++, FSW and Fair-DER++ sit close together on both fairness axes.** Deployed heads above 0.5 balanced accuracy on Bias in Bios. Dashed: empirical frontier. Error bars: ±1 SD over ten seeds.

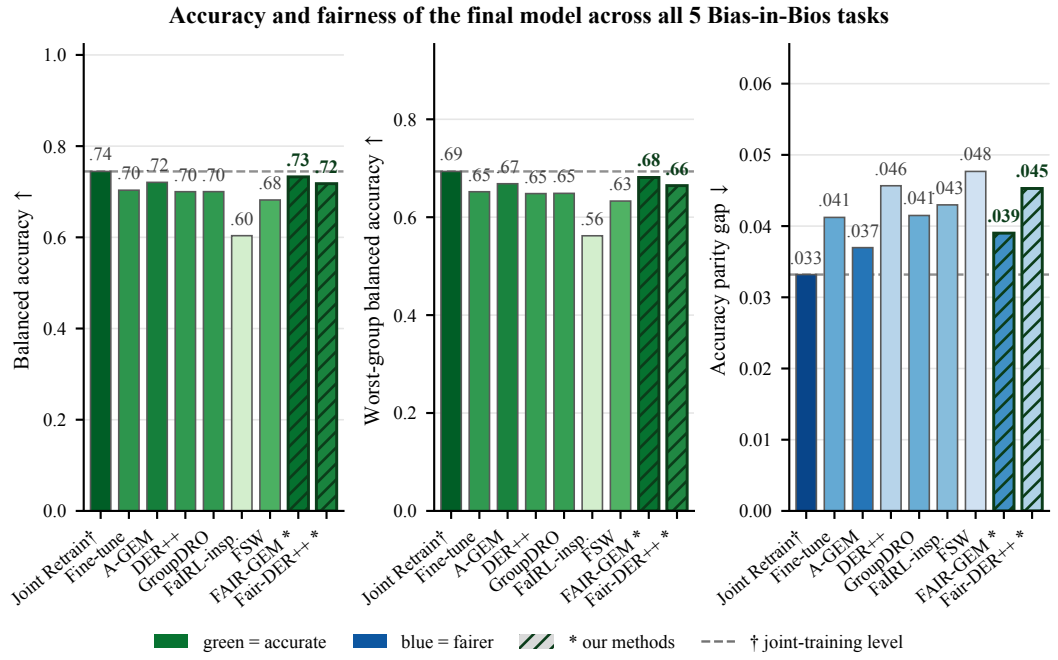


Figure 14: **With a refit head, every method except FaIRL-insp. comes within a few points of joint training.** Memory-refit results on Bias in Bios. Hatched: ours. Dashed: joint training.

Memory-refit results are additionally computed under a fairness-aware (Kamiran–Calders reweighed) head. This checks the readout weighting but does not remove the retained-data confound. Separately, the released shared-reference logistic-head protocol fits all methods on the same held-out validation examples. The NCM result in Table 6 is a different shared-data diagnostic: it computes each encoder's class centroids from all training examples and supplies the reported cross-method ranking.

G.27.1 Conditional-Surrogate Training Pilot

We piloted a label-conditioned training surrogate for Fair-DER++ on Bias in Bios under the identical tuned protocol (20 validation-only Optuna trials in a separate study), then probed the selected

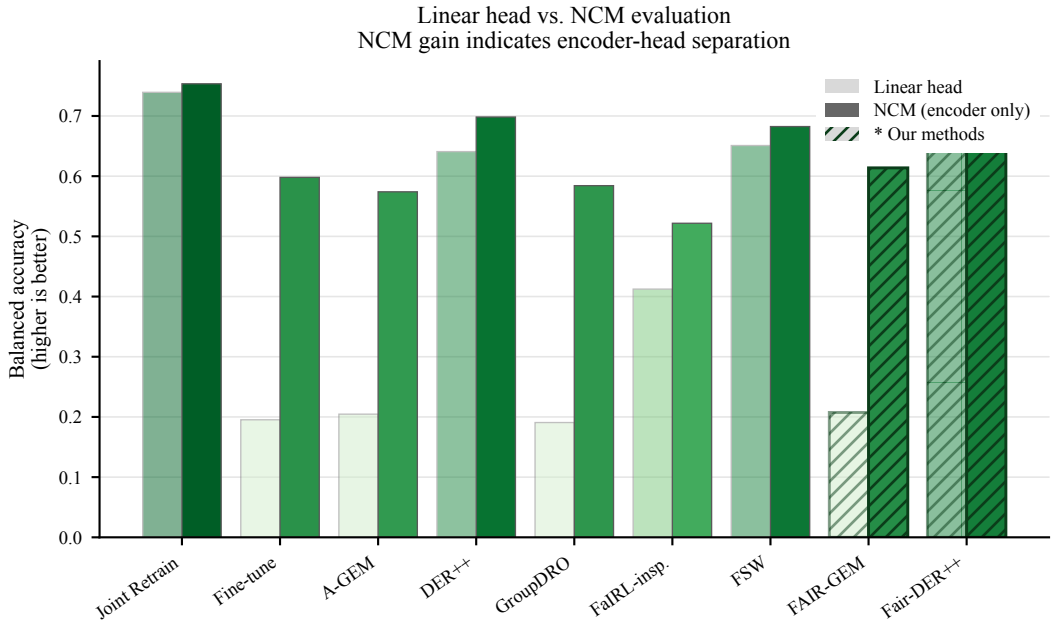


Figure 15: **A nearest-class-mean readout lifts every collapsed method far above its deployed head.** Deployed linear head (light) against nearest-class-mean classification on the frozen encoder (dark), which uses no trained head.

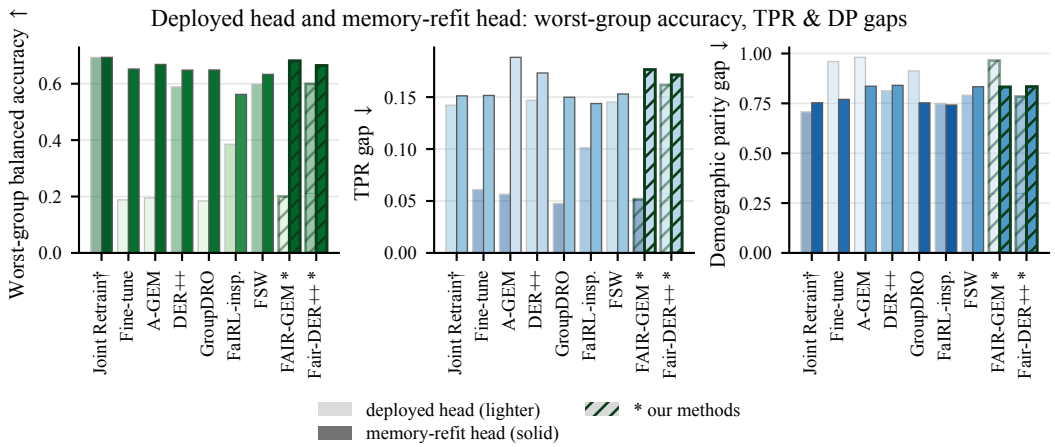


Figure 16: **Collapsed heads show small TPR gaps that refitting raises to the level of the working models.** Deployed (light) and memory-refit (solid) heads on three further metrics.

configurations on validation EO. Conditioning did not improve the composite objective (0.612 vs. 0.615), validation EO (0.160 vs. 0.154), or AP (0.079 vs. 0.066). We therefore retain the unconditional surrogate. A memory- or EMA-stabilized conditional estimator remains unevaluated (artifacts: `pilot_conditional_trials.csv`, `pilot_eo_probe.csv`).

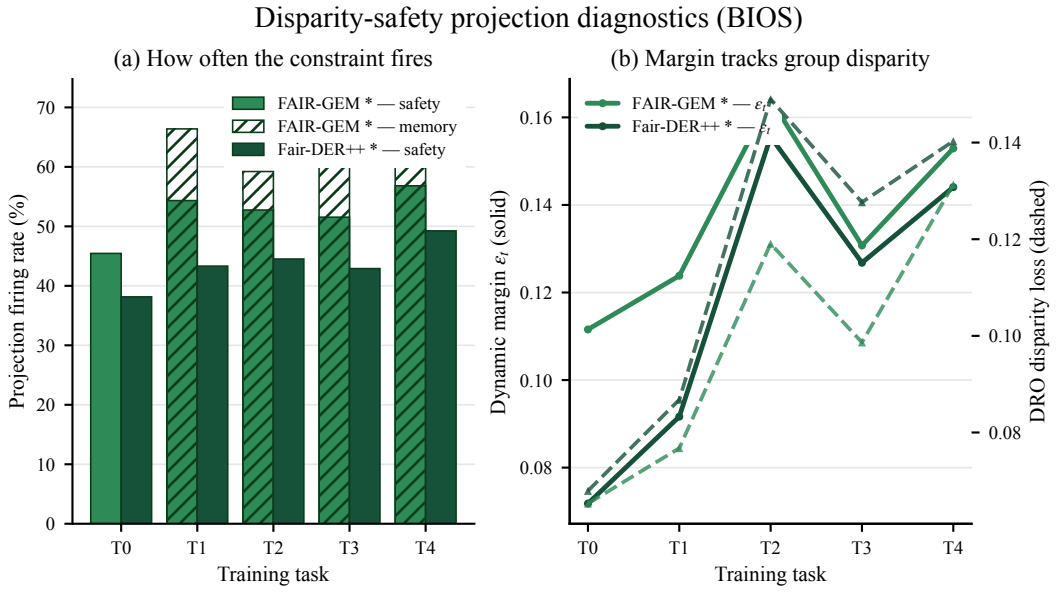


Figure 17: **The fairness constraint fires on roughly half of all steps, and its margin tracks the measured disparity.** (a) Projection firing rate per task. (b) Margin ε_t (solid) against group-loss disparity (dashed).

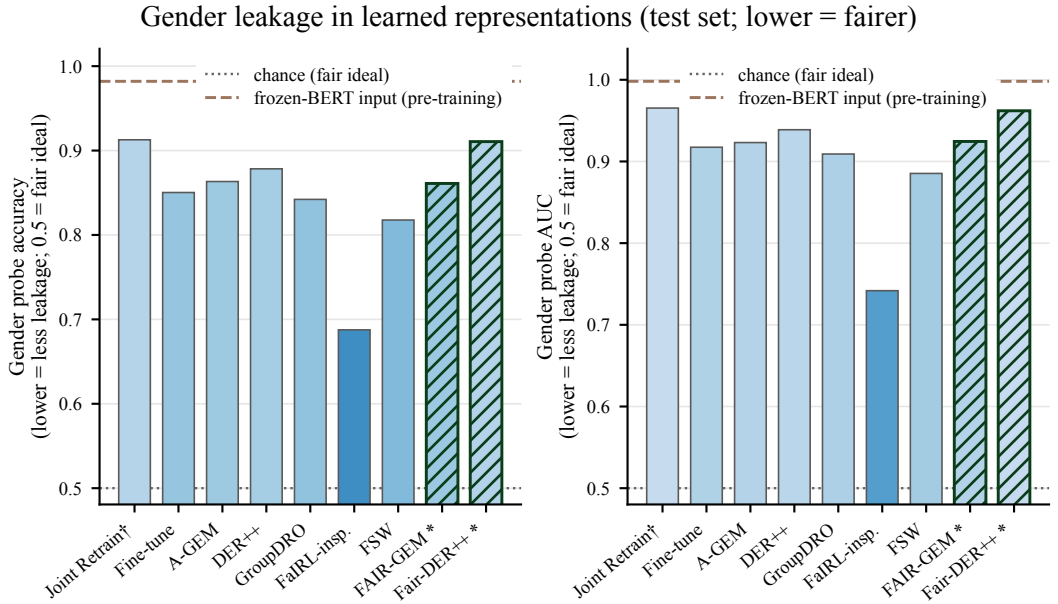


Figure 18: **Every encoder still encodes gender well above chance.** Accuracy (left) and AUC (right) of a gender probe on each final encoder. Dotted: chance. Dashed: pre-trained BERT. Mean over three seeds.

G.28 Surrogate-Alignment Scatter Plots

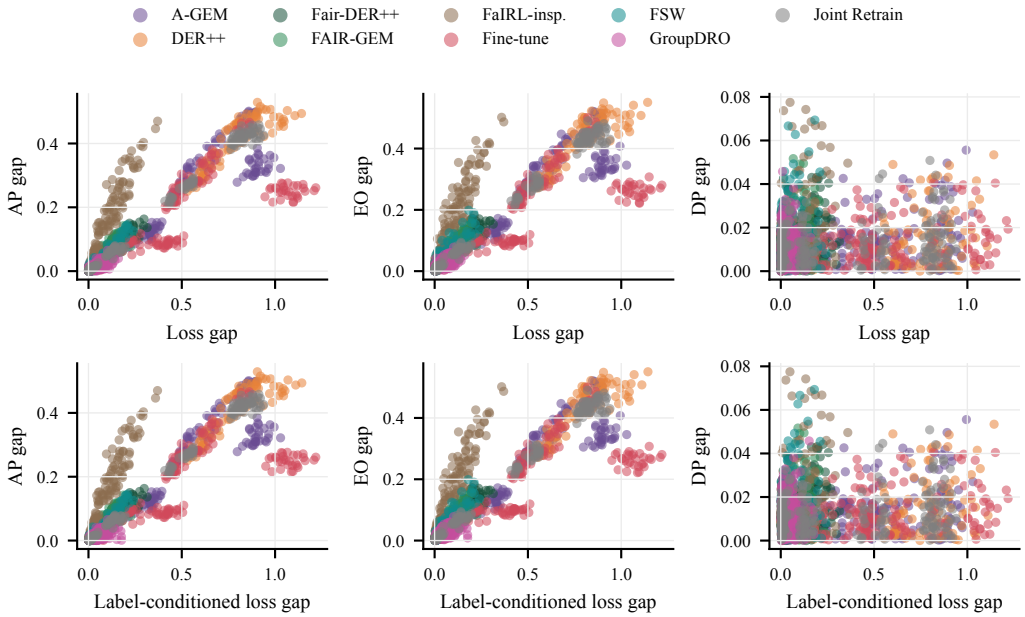


Figure 19: **On Spurious Shift the loss-gap surrogate tracks the accuracy-parity and equalized-odds gaps, but not demographic parity.** One point per method, seed and task. Top: unconditional loss gap. Bottom: label-conditioned.

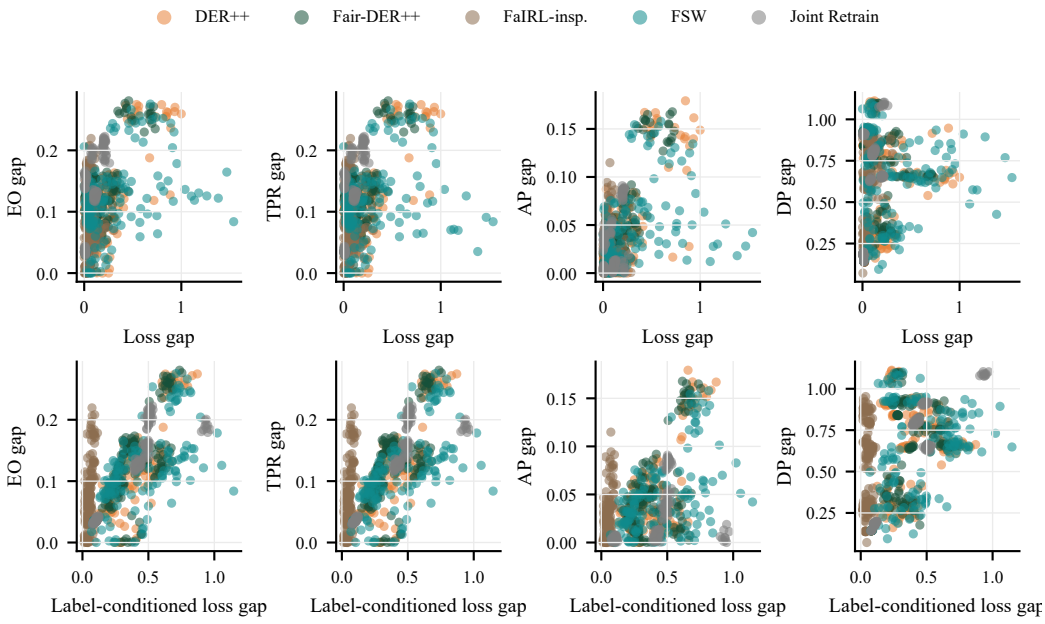


Figure 20: **On Bias in Bios, conditioning on the label strengthens the surrogate's link to the equalized-odds and TPR gaps.** Layout as in Figure 19, for the five non-collapsed methods.

G.29 Trade-off Sweep

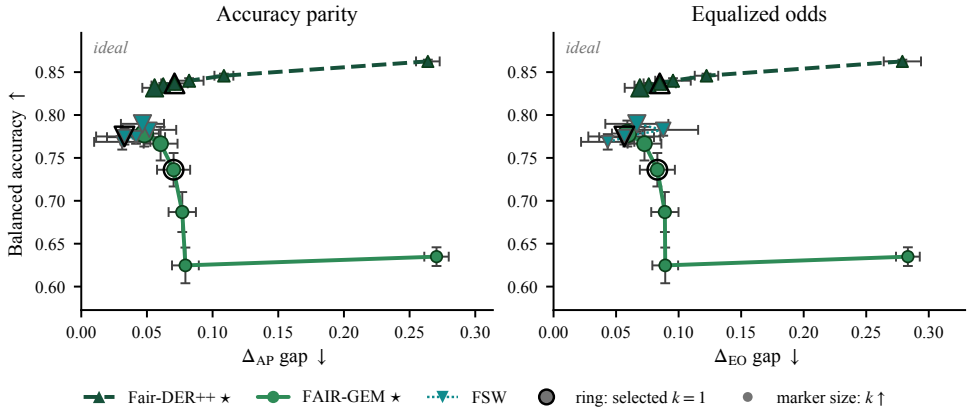


Figure 21: **Each method's fairness control traces its own accuracy–fairness curve on Spurious Shift.** Points scale the selected control by k (marker size), and the ring marks the selected $k = 1$. These come from a separate run, so $k = 1$ can differ slightly from Table 2. Error bars: ± 1 SD over ten seeds.

G.30 Extended Related Work

Continual learning. Catastrophic-forgetting methods fall into three families: *regularization* (EWC [Kirkpatrick et al., 2017], SI [Zenke et al., 2017]), *replay* (ER [Rolnick et al., 2019], DER++ [Buzzega et al., 2020]), and *gradient projection* (GEM [Lopez-Paz and Ranzato, 2017], A-GEM [Chaudhry et al., 2019], GPM [Saha et al., 2021]). Regularization penalizes parameter drift, replay preserves examples or logits, and gradient projection enforces local retention constraints. DER++ and A-GEM provide representative replay and constrained-update backbones, but neither targets demographic disparity. FAIR-GEM and Fair-DER++ add that missing constraint to A-GEM and DER++, respectively. Their projection geometry also connects to multi-objective gradient methods such as PCGrad and CAGrad [Yu et al., 2020, Liu et al., 2021], although our half-spaces encode explicit retention and disparity conditions rather than pairwise task-gradient conflict.

Fairness under distribution shift. GroupDRO [Sagawa et al., 2020] is the standard for group robustness, minimizing worst-group loss, but assumes i.i.d. data and group labels. Hashimoto et al. [2018] show that ERM amplifies representation disparity over repeated rounds, motivating fairness in sequential settings, and Jiang et al. [2023] derive a demographic-parity regularizer under simulated shift. These methods assume a single train–test split. Applied independently per task, they do not provide a retention mechanism. Influence-based reweighting instead estimates each training example's effect on validation utility and fairness [Li and Liu, 2022]. Like FSW, it intervenes through sample weights in a static training problem, whereas our constraints act on a replay backbone's update geometry throughout a task sequence.

Fairness in continual learning. FaIRL [Basu Roy Chowdhury and Chaturvedi, 2023] sustains fair representations via rate-distortion theory rather than prediction-level metrics like DP or EO. Recent approaches instead optimise metric-agnostic group fairness [Lian et al., 2025], suppress spurious correlations during continual learning [Capitani et al., 2025], or use contrastive attention for fair continual segmentation [Truong et al., 2025]. Ceccon et al. [2024] document fairness forgetting in medical imaging but propose no mitigation. FSW [Park et al., 2026] targets a related gradient-misalignment mechanism but intervenes on the data: it solves a linear program for per-sample weights so that the aggregate training gradient better satisfies a linearized fairness objective, discarding low-weight samples. Our methods instead start from the backbone's intended CL update and apply low-rank geometric corrections toward memory and disparity half-spaces. Both mechanisms use cross-entropy (CE) losses, but FSW uses label-conditioned class–group losses whereas our training constraint uses an unconditional worst–best group loss gap. We test both unconditional and label-conditioned variants empirically (Sec. D.2). We also compare against FSW directly under identical

replay budgets and the tuning protocols stated in Section B.1. We instantiate the constraint on two representative episodic-memory backbones and validate it on a controlled benchmark that isolates accuracy from fairness forgetting as well as on Bias in Bios.

Other projection families, and where our realized-step finding applies to them. A separate line of work projects updates to avoid interfering with earlier tasks rather than to satisfy a constraint: ROGO relaxes the orthogonality requirement to recover forward transfer [Yang et al., 2023], and CGP builds the projected subspace per class rather than per task [Chen et al., 2023]. We do not compare against them, because they optimise a different objective — retention, not disparity — and because our methods are instruments here rather than contributions. They are worth naming for one reason: our finding that the projected direction survives on a minority of applied updates under AdamW (Appendix G.20) is a property of the optimizer, not of what the projection was for, so any projection-family method reporting a per-step geometric guarantee inherits the same gap between the direction proved and the direction taken. None of the work above measures it.

Protocol-sensitivity findings elsewhere. The claim that evaluation protocol rather than method choice drives reported fairness differences is not ours alone. Tan et al. [2026] run a tuning-aware benchmark across vision and vision-language models and find that many debiasing methods do not reliably beat a well-tuned ERM baseline — the same shape as our result that a one-line loss penalty matches our projections' disparity on Spurious Shift (Appendix G.17), and the same warning that a comparison run at unequal tuning effort measures the effort. Ceccon et al. [2024] document fairness degradation under continual updates in a clinical setting without isolating a cause; the collapse artifact and the readout inversion reported here are candidate causes for contradictory results of that kind.

G.31 Baseline Details

We compare against six baselines and one full-data reference:

Full-data reference (oracle access). **Joint Retrain**: re-trains from scratch on all data seen so far. It is a joint-training reference with access to past data, not a continual learner.

CL baselines (no fairness mechanism). **Fine-tune**: sequential cross-entropy, no forgetting prevention (lower bound). **A-GEM** [Chaudhry et al., 2019]: projects the task gradient onto the half-space that does not increase past-task memory loss ($g_{\text{mem}}^{\top}\tilde{g} \geq 0$, 448 memory samples per step). **DER++** [Buzzega et al., 2020]: replay-based CL combining logit-distillation and label replay (loss in Appendix G.8).

Fairness baselines (no CL mechanism). **GroupDRO** [Sagawa et al., 2020]: exponentiated group weight updates (tuned η, C in Appendix G.26). It re-trains from scratch per task on Spurious Shift, but trains continually on BIOS because resetting would retain only the final task's classes.

Fair CL baselines. **FSW** [Park et al., 2026]: the closest prior method. It solves a linear program each epoch for per-sample weights of the current task so that linearized per-(class, group) losses equalize, drops samples below weight 0.05, and replays memory with coefficient τ. We reimplement the reference EO variant using last-layer gradient linearization, ℓ_2-normalized gradients, and the same LP structure with an open-source solver. For a controlled comparison, this adaptation shares our architecture, episodic-memory budget, replay batch size, and training schedule. It is not a reproduction of FSW's published Bias-in-Bios recipe, which uses 10 epochs, length-128 inputs, batch size 64, and SGD with momentum 0.9. Its hyperparameters (α, λ, τ) (and BERT learning rate on Bias in Bios) use the same Optuna protocol and trial budget as all other methods on both benchmarks. One protocol asymmetry remains and we state it: on Bias in Bios our FSW adaptation seeds its batch-order generator from the fixed configuration seed rather than the per-run seed, so its batch order is common to all ten runs while every other method's varies. Model initialisation, the validation split and memory sampling still vary per run, so FSW's reported spread is not zero, but it omits one source the other methods include. FSW enters no significance test here, and it does not beat ordinary replay on deployed-head EO, so the narrower spread does not favour our conclusions; correcting it requires re-running that arm. On Spurious Shift, tuning did not improve on the reference-grid configuration (tuned: 0.779 ± 0.012 BalAcc, 0.045 ± 0.033 AP gap; reference $(0.002, 0.5, 2.0)$:

0.778 ± 0.009, 0.037 ± 0.016). We nevertheless use the validation-selected configuration in every primary table and figure; the fixed configuration is reported only as a reference sensitivity result (`fsw_reference_summary.csv`). **FaIRL-insp.** [Basu Roy Chowdhury and Chaturvedi, 2023]: a scoped objective-level diagnostic, not a reproduction. It retains an MCR^2-style class coding-rate term, while our shared implementation uses discriminator-based encoder confusion rather than FaIRL's sensitive-attribute coding-rate and incremental- consistency objectives. On Spurious Shift it uses one discriminator step, unnormalized features, and ordinary replay. On Bias in Bios it uses cross-entropy discrimination, KL-to-uniform confusion, and no episodic memory. We therefore report it only under the FaIRL-insp. label.

All selected hyperparameters are listed in Appendix G.26.

NeurIPS Paper Checklist

1. **Claims.** [Yes] The abstract and introduction claim (i) that task-recency head collapse drives deployed-head fairness gaps below every functioning model's, so an unguarded table ranks the collapsed models first (Table 1, Figure 1), (ii) that refitting the head restores accuracy and the disparity with it, and every method then converges into one band (Figure 2, Sec. 4.2), and (iii) that no fairness-aware method improves deployed-head equalized odds over replay on Bias in Bios, while the same constraint cuts the gap on Spurious Shift (Sec. 4.3, Appendix C).
2. **Limitations.** [Yes] A dedicated Limitations paragraph covers the per-step-only scope of the theory, the AdamW realized-step gap with measured rates, multi-group/intersectional estimator sparsity, and the binary-attribute scope of the experiments.
3. **Theory, assumptions, proofs.** [Yes] The cancellation proposition states its expectation-level assumption with proof (Appendix G.6); the single-half-space projection is derived and the dual cyclic feasibility procedure is documented in Appendix G.7; Sec. E.1 states the exact/local/non-claim tiers explicitly, including what is not guaranteed.
4. **Reproducibility.** [Yes] Reported hyperparameters (Appendix G.26) match the frozen configurations shipped with the code. The Bias in Bios manifest covers all nine methods, and the Spurious Shift manifest includes a completed equal-budget FSW study. Runs are seeded, per-seed artifacts and audit scripts (tuning-version tripwires, memory-parity assertions, preflight config checks) are included in the supplement.
5. **Open access to data and code.** [Yes] Code, configs, per-seed results, and the Spurious Shift generator are available at `https://github.com/MlsBli/broken-model-reads-fairest`. Bias in Bios is a public benchmark auto-downloaded from its public host.
6. **Experimental details.** [Yes] Architectures, optimizers, schedules, memory budgets, tuning objective/budget, and data splits are specified in Sec. 4.1, Sec. 5.1, and Appendices G.9, G.10, G.11, G.26. Replay methods share the same configured 800-example memory capacity and equal realized allocations, asserted at runtime. Tuning budgets are disclosed separately for each benchmark.
7. **Statistical significance.** [Yes] All main results are mean $\pm$ sd over ten seeds (Spurious Shift) or ten seeds (Bias in Bios); summary and ablation figures report standard deviations, while heatmaps show cellwise seed variation. Correlation analyses report pooled method–seed cluster-bootstrap confidence intervals and method-specific/within-method statistics. Headline comparisons additionally carry exact two-sided paired permutation tests over all 2^n sign assignments, reported beside the attainable floor $2/2^n$ and beside Holm–Bonferroni values correcting for the 17 tests in the family (Appendix G.16); comparisons we do not test are phrased as observed differences with seed variation.
8. **Compute resources.** [Yes] Spurious Shift: CPU-only; recorded serial method times sum to 0.44 CPU-hours for the ten-seed main campaign, excluding ablations. That figure is environment-specific: an earlier environment recorded roughly two orders of magnitude more time for FSW alone (Appendix G.25). Bias in Bios: a single H100 80GB; recorded training wall-clock for the ten-seed sweep sums to 23.7 GPU-hours across all methods, with model selection (9 methods $\times$ 20 trials) and the post-hoc readout and probe stages additional. Per-method wall-clock tables ship with the code (Appendix G.25).
9. **Code of ethics.** [Yes] The research conforms to the NeurIPS Code of Ethics; it studies fairness preservation using public, de-identified data.
10. **Broader impacts.** [Yes] The work targets a positive societal impact (preserving group fairness under continual deployment); we note that group-level constraints require training-time access to sensitive attributes, which carries its own governance obligations, and that reduced average disparity does not certify individual-level fairness (Introduction and Conclusion/Limitations).
11. **Safeguards.** [N/A] No models or data with high misuse risk are released.
12. **Licenses for existing assets.** [Yes] Bias in Bios [De-Arteaga et al., 2019] is used via its public distribution with personal identifiers scrubbed (`hard_text`); BERT-base is Apache-2.0; both are cited.

13. **New assets.** [Yes] The Spurious Shift benchmark generator and all method implementations are released with documentation and the exact generating process (Appendix G.5).
14. **Crowdsourcing and human subjects.** [N/A] No crowdsourcing or human-subject research.
15. **IRB approvals.** [N/A] Not applicable.